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# Machine Learning Ensemble for Detecting Motor Insurance Fraud in Kenya

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# DECLARATION AND APPROVAL

We, the undersigned, declare that this proposal is our original work and has not been submitted for a degree in any other university.

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*Dedicated to our families and mentors,*
*whose unwavering support and encouragement*
*made this journey possible.*



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# Abstract

Motor insurance fraud imposes substantial financial costs on Kenya's insurance sector, yet existing detection systems identify fewer than 3% of estimated fraudulent claims. This study develops a three-modality stacked ensemble framework for detecting fraudulent motor insurance claims in Kenya. The framework integrates eXtreme Gradient Boosting (XGBoost), Long Short-Term Memory networks (LSTM), and Graph Neural Networks (GNN), and is rigorously evaluated and validated on both benchmark and Kenyan-specific datasets.

The ensemble was trained on a 15,207-record Kaggle insurance dataset after hybrid SMOTE + ENN resampling and validated out-of-sample on a 3,200-record simulated Kenyan dataset. A logistic regression meta-learner combined out-of-fold probability predictions from the three base models.

The stacked ensemble achieved an F1-Score of 85.34%, AUC-PR of 0.902, and Recall of 86.44% on the held-out test set, representing a 9.67 percentage point improvement over the best individual model (XGBoost). All improvements were statistically significant (Wilcoxon signed-rank, $p < 0.01$). On the Kenyan validation set, the ensemble maintained F1 = 82.57% and AUC-PR = 0.8743, demonstrating robust domain generalisability. Ablation studies confirmed each modality contributes uniquely: removing XGBoost incurs $-6.91$ pp F1; removing the GNN costs $-4.10$ pp, particularly for syndicate fraud rings.

The confidence-scoring framework concentrates fraud prevalence to 67.8% in the "Investigation Required" tier versus 13.2% in the "High Confidence" tier, enabling rational investigator triage. The study bridges a documented gap between global ensemble best-practices and the Kenyan motor insurance fraud landscape, providing a deployable, interpretable detection pipeline for Kenyan insurers.

# List of Acronyms

| | |
|---|---|
| AI | Artificial Intelligence |
| AUC-PR | Area Under the Precision-Recall Curve |
| AUC-ROC | Area Under the Receiver Operating Characteristic Curve |
| CV | Cross-Validation |
| ENN | Edited Nearest Neighbour |
| GAT | Graph Attention Network |
| GCN | Graph Convolutional Network |
| GNN | Graph Neural Network |
| IRA | Insurance Regulatory Authority |
| KNN | K-Nearest Neighbours |
| LIME | Local Interpretable Model-agnostic Explanations |
| LSTM | Long Short-Term Memory |
| ML | Machine Learning |
| RF | Random Forest |
| SHAP | SHapley Additive exPlanations |
| SMOTE | Synthetic Minority Over-sampling Technique |
| SVM | Support Vector Machine |
| XAI | Explainable Artificial Intelligence |
| XGBoost | eXtreme Gradient Boosting |

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Chapter 1

# Introduction

## 1.1 Background of the Study

Insurance fraud is a deliberate act of deception committed against or by an insurance company or agent for the purpose of financial gain. A fraudulent motor insurance claim is a situation where the insured seeks financial gain by either filing forged documents or fabricating an accident or motor theft (Maina, Moso, & Gikunda, 2023). In Kenya, insurance plays a critical role in the economy. According to (Team, 2025), Kenya's insurance industry had a gross written premium of KES 395.3 bn in 2024, a 9.4% increment from 2023, showing steady growth of the premium. Motor insurance, a major segment, allows vehicle owners to transfer financial risk to insurers in exchange for premium payments. However, fraudulent claims, where policyholders seek illegitimate gain by fabricating accidents or submitting forged documents,are a rampant issue.

It is estimated that while 10% (Itri, Mohamed, Mohammed, & Omar, 2019) of all claims are fraudulent, fewer than 3% (Kowshalya & Nandhini, 2018) are legally pursued. For this reason, there is a need for efficient ways to detect fraudulent claims reported to insurance companies that will reduce loss adjustment expenses. If an insurance company is able to identify perpetrators of fraud and take necessary action, the customer satisfaction rate will increase and loss adjustment expenses will be reduced. Improved customer satisfaction, driven by faster legitimate claim processing, may reduce the adversarial dynamic that incentivises fraudulent claiming behaviour. Reduced loss adjustment increases the profit made by insurance companies and reduces the premium rating, thus making insurance policies more affordable.

Machine learning offers a powerful toolset for identifying patterns, anomalies, and intricate relationships within large datasets that may be indicative of fraudulent behaviour. By leveraging historical claims data and a wide range of features, such as policyholder information, vehicle details, and incident circumstances, machine learning models can be trained to accurately classify claims as legitimate or fraudulent (Z. Zhang, 2024). While existing research has made significant strides in applying individual machine learning algorithms to insurance fraud detection, a critical limitation persists. Past studies have predominantly focused on identifying the single best-performing model. Machine learning, particularly ensemble methods that combine multiple models, has demonstrated strong promise in enhancing detection accuracy and

reducing financial losses. Such approaches can analyse complex patterns and relationships in claims data, providing a more robust mechanism for identifying fraud.

This research proposes a sophisticated ensemble model that leverages the unique capabilities of three distinct machine learning algorithms: eXtreme Gradient Boosting (XGBoost), Long Short-Term Memory networks (LSTM), and Graph Neural Networks (GNNs). The model is designed to capitalise on their complementary strengths to create a more robust and accurate fraud detection system. XGBoost provides a powerful, interpretable baseline on structured claims data. LSTM models analyse the temporal sequences of a claimant's history to identify behavioural patterns. GNNs uncover complex, collaborative fraud rings by modelling relationships between entities such as claimants, garages, and agents. The weaknesses of each individual model are mitigated within the ensemble framework. By synthesising the strengths of these algorithms, the proposed system aims to provide a multi-faceted defence against insurance fraud, enhancing detection accuracy and operational efficiency for insurers in Kenya.

## 1.2 Statement of the Problem

Although the Kenyan insurance industry makes a substantial contribution to the economy, it incurs huge financial losses due to fraudulent motor insurance claims that remain undetected. Current detection systems remain inadequate: only a small proportion, fewer than 3%, of the estimated 10% of fraudulent claims are formally investigated (Kowshalya & Nandhini, 2018). Consequently, insurers incur unnecessary Loss Adjustment Expenses, honest policyholders face elevated premiums, and market-wide trust in the insurance sector is eroded.

A fundamental limitation escalating this challenge is the reliance on simplistic or single-model machine learning approaches for fraud detection. These approaches are inherently flawed; each individual model has specific limitations. Certain models work well with structured data, while being unable to contextualise textual cues or identify complex relational networks that are typical of an organised fraud ring. This gap in functionality leaves insurers vulnerable to collaborative fraudulent schemes, necessitating the development of more comprehensive detection frameworks.

This research answers that need by systematically developing an integrated ensemble model. The integrated ensemble model incorporates the best of what XGBoost does with structured data, what Long Short-Term Memory (LSTM) networks do with textual data, and what Graph Neural Networks (GNN) do to find relational networks. This comprehensive approach is essential to significantly improve detection accuracy, reduce financial losses, and enhance the sustainability of the Kenyan motor insurance market.

## 1.3 Objectives of the Study

### 1.3.1 Main Objective

To build and evaluate a high-performance ensemble model that integrates XGBoost, LSTM, and GNN to detect fraudulent claims in Kenya's motor insurance sector.

### 1.3.2 Specific Objectives

1. To develop a stacked ensemble model integrating XGBoost, LSTM, and Graph Neural Networks for motor insurance fraud detection.
2. To evaluate and compare the performance of the ensemble model against individual base models using classification metrics such as Accuracy, Precision, Recall, F1-score, and AUC.
3. To validate the generalisability of the proposed model using a Kenyan motor insurance dataset through contextual (proxy-based) validation.

## 1.4 Justification

This research is urgently needed in view of rampant financial losses and market inefficiencies stemming from undetected motor insurance fraud in Kenya. The vast difference between the estimated high prevalence of fraudulent claims (10%) (Itri et al., 2019) and the low proportion examined (fewer than 3%) (Kowshalya & Nandhini, 2018) leaves much room for loss adjustment costs at the insurer, excessive premiums paid by sincere policyholders, and eroded trust in the market.

The underlying cause resides in the inefficacy of current detection systems, which employ single-model machine learning techniques. These models have inherent limitations; a model expert in structured data will likely not be able to handle textual claims narratives or recognise the rich relational graphs inherent in structured rings of fraud. This technical gap provides a critical weakness.

Accordingly, this study endeavours to bridge this gap through the development of an elaborate, integrated ensemble model. The framework aggregates XGBoost, LSTM, and Graph Neural Networks' complementary capacities in order to provide a robust solution something which dominant methods do not offer and directly targets this multi-faceted insurance fraud entity.

## 1.5 Significance of the Study

This study will be of great importance to numerous players in the Kenyan insurance industry. Insurers can have a revolutionary instrument for fighting fraud in the form of the proposed ensemble model, which can substantially reduce annual fraud-related losses through improved detection rates and fewer false positives. This will directly boost profitability and operational effectiveness. More accurate fraud detection can benefit policyholders through lower premiums and quicker processing of valid claims, fostering greater trust and satisfaction. Regulators, including the Insurance Regulatory Authority (IRA), will have at their disposal a data-driven model to keep abreast of industry trends and fortify compliance. The study offers academically a new, scalable machine learning architecture bespoke to emerging markets, enriching knowledge in fraud analytics and offering an adaptable paradigm to other areas of the insurance business beyond motor. This study contributes to Kenya's sustainable insurance sector through fostering technological innovation and financial integrity.

## 1.6 Research Hypothesis

$H_0$**:** There is no statistically significant difference in fraud detection performance between the proposed ensemble model and the standalone models.

$H_1$**:** The proposed ensemble model achieves statistically significant improvement in fraud detection performance compared to standalone models.

## 1.7 Research Questions

1. How does the predictive performance of the ensemble model (XGBoost + LSTM + GNN) compare to that of individual machine learning models (XGBoost, LSTM, GNN) in detecting motor insurance fraud in Kenya?

2. To what extent does the integration of multiple data modalities (structured, textual, relational) improve fraud detection accuracy and reduce false positives compared to single-modality approaches?

3. How can the proposed ensemble model be operationalised within existing Kenyan insurance claim processing workflows to maximise efficiency and scalability?

Chapter 2

# Literature Review

## 2.1 Motor Insurance Fraud Detection

Motor insurance is a popular insurance product in Kenya, protecting vehicle owners against financial losses or damages to the insured vehicle. Fraudulent claims remain a persistent challenge in the insurance industry, as policyholders or organised rings seek unfair financial gain at the insurer's expense. Fraud in motor insurance can take several forms, including inventing accidents, inflating damages, misrepresenting facts, and conspiring with third parties (Njeru, 2022). Global studies suggest that 10% of all claims are fraudulent, yet only a tiny number (fewer than 3%), (Maina et al., 2023) are reported, highlighting a significant gap in detection capability.

In 2020, fraudulent claims were anticipated to account for 25–35% of total claims, giving rise to underwriting losses exceeding KShs 6 billion (Njeru, 2022). The importance of identifying motor insurance fraud is highlighted by its economic and social impact. At the macro level, fraud costs insurers billions annually in unjustified payments, with European insurers losing an estimated €2.5 billion in 2017 alone (Schrijver, Sarmah, & El-Hajj, 2024). In Kenya, the Insurance Regulatory Authority (IRA) reported that motor insurance accounts for the largest share of non-life premiums but also records the highest underwriting losses due to fraud-related costs. This liability not only reduces the insurer's profitability but also leads to an escalation of premiums paid by honest policyholders, destroys consumer confidence in insurers (A. M. Wambu, 2025), and could at extreme levels cause systemic damage to insurers.

The conventional fraud detection methods that depend on manual probing and rule-based systems have proved unsuccessful. These are slow-moving, resource-intensive, and incapable of processing the ever-changing strategies of fraudsters. As noted by (Schrijver et al., 2024), fraud is not self-exposing and often forms within cooperative networks, making early identification very difficult. Machine learning (ML) strategies such as Decision Trees, Random Forests, XGBoost, LSTMs, and Graph Neural Networks have improved detection accuracy, leveraging large amounts of structured and unstructured claims data. Though ML models can identify subtle threats and anomalies that conventional systems miss, ensembling these models demonstrates increased ability to detect both individual and collective fraud patterns (A. M. Wambu, 2025). In

Kenyan case law, Njeru (Njeru, 2022) and Wambu (A. M. Wambu, 2025) suggest that ensemble models and hybrid feature selection approaches beat single algorithms with respect to recall, accuracy, and false positive reduction.

Hence, fraud detection mechanisms are essential not only for insurers' financial protection but also for fairness and affordability towards policyholders, building trust in the insurance industry, and supporting the longterm sustainability of the motor insurance sector.

## 2.2 Fraud and Traditional Detection Methods

The roots of fraudulent behaviour in insurance can be understood through several theoretical models. Most referred to is Cressey's Fraud Triangle Theory, which asserts that fraud emerges where three elements converge: pressure (financial or social incentive), opportunity (weak or susceptible systems), and rationalisation (individual justification of unethical behaviours) (A. M. Wambu, 2025). The Fraud Diamond extended this by adding the element of ability. Agency Theory accounts for fraud as an outcome of conflicting interests between policyholders (agents) and insurers (principals), (Njeru, 2022) where asymmetric information creates an incentive to defraud .

Traditional fraud detection methods rely predominantly on rule-based systems and labour-intensive investigations. Investigators manually review cases, interview claimants, and verify auxiliary documents. Although occasionally successful, such methods are tedious, labour-heavy, and not scalable for high-volume cases typical of motor insurance (Schrijver et al., 2024). Rule systems that call upon pre-set parameters incorporated an element of automation, but they are rigid they do poorly when fraudsters devise new tricks outside established patterns (Schrijver et al., 2024) and typically produce high false-positive ratios.

Kenyan insurers, like their global counterparts, have traditionally relied on manual investigations and rule-based systems. Reports state that while an estimated 10% of motor insurance cases are fraudulent, fewer than 3% are legally prosecuted, illustrating the shortcomings of traditional approaches (Maina et al., 2023). The ineffectiveness, excessive cost, and inability to identify new fraud trends have made room for more sophisticated, data-driven approaches.

## 2.3 How Machine Learning Has Been Applied in Fraud Detection Globally

Ensemble learning combining multiple base learners to develop a stronger meta-model became the predominant method of fraud detection because of its promise of improving predictive acumen, robustness, and generalisability over single classifiers. Bagging (bootstrap aggregation), boosting (sequential error-correcting learners), and stacking (meta-learning over base learner

outputs) are the three ensemble strategies most widely applied in practice and experiments. Review articles routinely report that ensembles outperform single models consistently (Ming, Li, & Chen, 2024; Khalid, Ahmed, & Khan, 2024; Eteng, Okon, & Bassey, 2025; Scrivano, 2024), provided attentive pre-processing correcting for class imbalance and feature selection is applied.

### 2.3.1 Bagging

Bagging reduces variance by training a multitude of copies of a base model on bootstrap resamples of the training set and then aggregating their predictions. Random Forest (RF), a decision-tree based ensemble typical of the bagging approach, is often described as a suitable baseline for fraud problems ,(Ming et al., 2024; Eteng et al., 2025) it is stable with respect to noisy features, handles high-dimensional data well, and frequently achieves competitive precision/recall with minimal hyperparameter tuning .

### 2.3.2 Boosting

Boosting algorithms (AdaBoost, Gradient Boosting, XGBoost, LightGBM, CatBoost) iteratively focus learners' attention on previously misclassified instances and are effective at finding hard to detect fraud patterns. Insurance and credit card papers document superior results from gradient boosting variants (Ming et al., 2024; Khalid et al., 2024), especially after feature selection and hybrid resampling such as SMOTE.

### 2.3.3 Stacking

Stacking (stacked generalisation) trains a meta-learner on the outputs of multiple base classifiers and is frequently used to integrate complementary model strengths linear models, tree ensembles, SVMs, and neural nets. Recent applied work demonstrates that stacking is particularly effective for very imbalanced fraud datasets, (Eteng et al., 2025; Y. Zhang, Liu, & Chen, 2022) stacked ensembles combining tree-based models, SVM, KNN, Naive Bayes, and a logistic regression meta-classifier demonstrate large recall/precision gains when combined with resampling.

**Table 2.1.** Comparison of Ensemble Methods in Fraud Detection

| Method | Strengths | Weaknesses | Typical Libraries & Key Hyperparameters | Recommended Resampling |
|---|---|---|---|---|
| **Bagging** | • Reduces variance, robust to noise<br>• Handles high-dimensional data<br>• Strong baseline (Ming et al., 2024; Eteng et al., 2025) | • Can overfit with deep trees<br>• Less effective with highly imbalanced data (Scrivano, 2024) | • `scikit-learn`, `R randomForest`<br>• `n_estimators`, `max_depth`, `max_features` | • Undersampling<br>• Hybrid SMOTE+ENN (Eteng et al., 2025) |
| **Boosting** | • Captures complex patterns<br>• Focuses on hard cases<br>• High accuracy (Ming et al., 2024; Khalid et al., 2024) | • Sensitive to noise/outliers<br>• Risk of overfitting without tuning<br>• Computationally expensive (Scrivano, 2024) | • `XGBoost`, `LightGBM`, `CatBoost`<br>• `learning_rate`, `n_estimators`, `max_depth`, `subsample` | • SMOTE and hybrid SMOTE+Tomek<br>• Cost-sensitive boosting (Khalid et al., 2024) |
| **Stacking** | • Combines diverse learners<br>• Exploits complementary strengths<br>• Strong recall/precision (Eteng et al., 2025; Y. Zhang et al., 2022) | • More complex to train<br>• Overfitting risk without careful meta-learner design<br>• Computationally heavier (Scrivano, 2024) | • `mlxtend`, `scikit-learn`<br>• Base: RF, SVM, KNN, NB<br>• Meta: Logistic Regression, GBM | • SMOTE + undersampling<br>• CV essential to avoid overfitting (Eteng et al., 2025; Y. Zhang et al., 2022) |

### 2.3.4 Global Application to Fraud Detection

#### *2.3.4.1 Handling Class Imbalance*

Fraud datasets are severely imbalanced (frauds far fewer than legitimate claims). Past studies overwhelmingly support that ensembles be coupled with data-level methods: oversampling (SMOTE), undersampling, and hybrid methods (SMOTE + ENN). Experimental comparisons show that the same ensemble architecture will produce wildly varying results depending on resampling strategy, (Eteng et al., 2025; Khalid et al., 2024) hybrid resampling tends towards an optimal recall/false-positive trade-off .

#### *2.3.4.2 Feature Selection and Interpretability*

Ensembles benefit substantially from feature engineering (deriving temporal, relational, and aggregated features) and from feature selection (filter, wrapper, embedded methods). Because boosting and stacking models can be opaque, recent work (Ming et al., 2024; Scrivano, 2024) incorporates explainable AI tools (SHAP, feature importance plots) to support regulatory compliance and investigator trust.

#### *2.3.4.3 Global Lessons for Motor Insurance Fraud*

The global literature consistently shows that ensemble methods (bagging, boosting, stacking) outperform individual models in fraud detection tasks. Translating these best practices to the Kenyan motor insurance context implies developing pipelines that combine ensemble classifiers with resampling and robust feature engineering, while also applying explainability tools (P. Wambu, 2025) to ensure transparency for regulators and insurers.

## 2.4 Summary of Empirical Studies

Empirical studies of fraud detection consistently demonstrate that machine learning ensemble methods supersede established statistical-based and single-model approaches.

### 2.4.1 Bagging-Based Methods

Bagging, particularly Random Forests (RF), has been widely studied for insurance fraud detection and financial applications. Papers (e.g., (Ming et al., 2024; Njeru, 2022)) report that RF achieves good baseline performance because of its robustness against noise and high-dimensional data. As an example, (Njeru, 2022) employed RF in detecting fraudulent motor insurance claims in Kenya, concluding that ensemble models were superior to logistic regression and decision trees. However, RF's performance worsens significantly in extremely imbalanced datasets characteristic of fraud detection.

### 2.4.2 Boosting-Based Methods

Boosting algorithms such as AdaBoost, XGBoost, LightGBM, and CatBoost have shown high effectiveness in uncovering fraudulent patterns in structured insurance data. (Khalid et al., 2024; Scrivano, 2024) demonstrate that gradient boosting variants consistently achieve higher recall and precision than bagging when paired with resampling methods such as SMOTE or hybrid oversampling undersampling. While boosting is more computationally intensive and overfits if not fine-tuned correctly, empirical evidence shows it is extremely effective at uncovering rare fraudulent cases in credit card and insurance domains.

### 2.4.3 Stacking Approaches

Stacked ensembles that combine many different base learners using a meta-classifier have been applied to very imbalanced fraud datasets. Empirical experiments (e.g., (Eteng et al., 2025; Z. Zhang, 2024)) show that integrating decision trees, SVMs, $k$-NNs, and Naive Bayes significantly outperforms individual classifiers. A logistic regression or gradient boosting meta-learner did very well at improving recall while maintaining false positive rates at an acceptable level.

*Comparative and review studies.* Meta-analyses and systematic reviews verify the empirical consensus. Ashtiani and Raahemi (2021) summarised uses of ML and data mining for financial fraud detection, concluding that ensembles are among the best performing methods for handling skewed fraud data. Similarly, (Schrijver et al., 2024) conducted an extensive literature review of car insurance fraud detection by data mining and found that ensemble models, particularly tree-based models, dominate empirical studies due to their credibility and scalability.

#### *2.4.3.1 Graph Neural Networks for Relational Fraud Detection*

While bagging, boosting, and stacking ensembles have dominated the fraud detection literature, Graph Neural Networks (GNNs) represent an emerging paradigm specifically suited to detecting organised fraud rings that operate through collusive networks. Unlike feature-based models that treat each claim as an independent observation, GNNs model the relationships between entities policyholders, garages, agents, and assessors as a graph, (Liu, Wang, & Chen, 2022) enabling the detection of anomalous connection patterns invisible to conventional classifiers .

Recent applications of GNNs to financial fraud detection have demonstrated their unique value. (Wang & Liu, 2021) applied Graph Attention Networks (GAT) to health insurance claims, identifying fraudulent syndicates with 23% higher recall than XGBoost alone by detecting densely connected subgraphs of providers and patients. (Dou & Liu, 2020) used Graph Convolutional Networks (GCN) for credit card fraud detection, demonstrating that relational features capturing transaction co-occurrence patterns improved fraud recall by 18% over feature-based baselines. In the automotive domain, (Li & Zhang, 2023) constructed a claims graph from Chinese motor

insurance data and showed that GNN-based community detection identified 87% of known fraud rings compared to 41% for rule-based systems.

However, GNN applications to insurance fraud remain scarce, and no published study has applied GNNs to motor insurance fraud detection in an African market context. This gap is particularly significant given the prevalence of organised fraud rings in the Kenyan motor insurance sector (Njeru, 2022). The present study addresses this gap by integrating a GNN component specifically designed to detect syndicate coordination patterns within the Kenyan claims network.

### 2.4.4 Contextual Evidence from Kenya and Africa

Although most empirical work focuses on Europe, North America, and Asia, there is emerging evidence from Africa validating these findings at the local level. (Njeru, 2022) employed ML ensemble models for the Kenyan motor insurance market and found them more effective than traditional approaches. Nevertheless, gaps exist concerning large-scale, sub-Saharan Africa real-world deployments, providing potential for additional empirical testing within the Kenyan insurance market.

### 2.4.5 Synthesis

Taken collectively, empirical evidence strongly favours the adoption of ensemble fraud detection approaches. Bagging methods like Random Forests provide robustness and interpretability, boosting methods like XGBoost and LightGBM provide high recall for rare instances of fraud, and stacking offers state of the art efficiency by combining complementary learners. The empirical evidence is that ensembles outperform individual models significantly. Issues such as class imbalance, high computational expense, and non-interpretability persist. Evidence from Kenya is minimal but rising, emphasising the requirement for local insurance fraud studies specifically adapted to local features.

## 2.5 Research Gaps in Machine Learning for Motor Insurance Fraud Detection in Kenya

Despite significant progress in applying machine learning to fraud detection globally, gaps persist in its application to motor insurance fraud in Kenya and other emerging nations.

### 2.5.1 Limited Local Empirical Validation

Most empirical fraud detection papers utilising ML focus on America and Asia (Ashtiani & Raahemi, 2021; Schrijver et al., 2024; Khalid et al., 2024). Few studies, e.g., (Njeru, 2022), applied machine learning models to the Kenyan motor insurance market, but these are small

scale or theoretical in scope. No large scale, real-world deployments have been verified on Kenyan insurer datasets.

### 2.5.2 Class Imbalance Handling Underexplored in Kenya

International reviews highlight that fraud datasets are very imbalanced. Emerging resampling strategies such as SMOTE variants, hybrid oversampling, undersampling approaches, and cost-sensitive learning are highly recommended (Eteng et al., 2025; Z. Zhang, 2024; Ming et al., 2024). However, most Kenyan-directed studies rely on traditional oversampling or simple class weighting, leaving a gap in studying advanced imbalance treatments locally.

### 2.5.3 Excessive Model Dependence

Though global analyses prove ensemble strategies consistently outperform single classifiers (Ming et al., 2024), Kenyan studies (e.g., (Njeru, 2022)) have only evaluated decision trees, logistic regression, or Random Forests in isolation. A void exists for comparing and optimising state-of-the-art ensemble methods for motor insurance fraud in Kenya.

### 2.5.4 Feature Engineering and Contextual Variables

Global studies focus on temporal feature engineering, relational features (claimants' relationships with garages and cars), and aggregate behavioural features as the basis of improving ensemble performance (Scrivano, 2024; Schrijver et al., 2024). Kenyan datasets are rarely engineered beyond the initial claim attribute level, creating a gap in designing contextualised local fraud features such as staged crashes, syndicate damage, and pseudo-documentation.

### 2.5.5 Interpretability and Regulatory Trust

Insurance fraud detection requires not only efficiency but also transparency for regulators and insurers. International research increasingly integrates explainable AI tools like SHAP and LIME into ensemble models (Scrivano, 2024). Local research has not yet integrated interpretability frameworks, creating a gap in developing explainable ensemble models that meet regulatory and ethical demands in the Kenyan insurance market.

### 2.5.6 Integration with Operational Systems

Worldwide, fraud detection models are typically evaluated within near real-time processes or as part of integrated claims management systems (Scrivano, 2024). In Kenya, existing work remains predominantly academic with no end-to-end fraud detection pipelines implemented in insurer operations.

### 2.5.7 Scarcity of Comparative Benchmarking

Several international studies compare multiple ML algorithms against common datasets, allowing robust comparison (Ashtiani & Raahemi, 2021; Z. Zhang, 2024). Kenyan studies do not tend to include systematic benchmarking across ensemble, boosting, and deep learning approaches, providing room for evidence-based advisory for insurers on the most suitable ML methods locally.

### 2.5.8 Synthesis

Comprehensively, empirical research of ML for Kenyan motor insurance fraud detection is limited, with potential for large-scale empirical verification, addressing class imbalance issues, employment of high-level ensembles, feature engineering that accounts for local data distribution context, explainability, organisational integration, and systematic comparison.

## 2.6 Summary of Literature Review

The literature review demonstrates that machine learning ensemble techniques bagging, boosting, and stacking significantly dominate single-model and conventional statistical methodologies in fraud identification across varied applications, such as credit cards, banking, and insurance. Empirical studies confirm their enhanced capability to identify infrequent fraudulent behaviours, handle high-dimensional data, and gain enhanced recall and precision, specifically when accompanied by resampling and feature engineering methodologies. The ensembles are further supported by systematic reviews (Ashtiani & Raahemi, 2021), as the most effective and scalable methods to identify fraud.

Even with this global advancement, analysis within the Kenyan motor insurance market is still limited. Locally based research (Njeru, 2022) gives initial support for machine learning models but is frustrated by small-scale datasets, lack of high-end ensembles, and inadequate focus on class imbalance, feature engineering, interpretability, and operational integration. As such, there is a specific research gap in the adaptation and empirical testing of sophisticated ensemble techniques under the peculiar fraud patterns and regulatory regime of Kenya.

Chapter 3

# Research Methodology

## 3.1 Research Design

This study employs a comparative quantitative research design to develop, train, and evaluate machine learning models for detecting fraudulent motor insurance claims. The design follows a systematic model comparison framework where the predictive performance of individual models (XGBoost, LSTM, and GNN) is rigorously compared against a proposed ensemble model that integrates all three approaches.

### 3.1.1 Design Rationale

The comparative design was selected for several key reasons:

- *Model Evaluation Focus*: The primary research question involves comparing model performance rather than testing causal hypotheses.
- *Real-world Relevance*: Uses authentic insurance claims data representing actual fraud patterns in the Kenyan context.
- *Methodological Rigor*: Enables controlled comparison through standardised evaluation metrics and cross-validation procedures.
- *Practical Applicability*: Provides evidence-based guidance for insurers on optimal fraud detection approaches.

### 3.1.2 Research Phases

The research is organised into four main phases:

1. *Data Acquisition and Preprocessing*: Collection and preparation of multimodal insurance claims data.
2. *Base Model Development*: Independent training and optimisation of XGBoost, LSTM, and GNN models.
3. *Ensemble Construction*: Integration of base models using stacked generalisation.

4. *Performance Evaluation*: Rigorous comparison using imbalanced classification metrics and statistical testing.

This design supports both exploratory analysis (identifying predictive features from multimodal data) and confirmatory analysis (statistically comparing model performances). The ensemble approach ensures methodological robustness by integrating diverse algorithmic strengths boosting for structured patterns, recurrent networks for temporal dynamics, and graph networks for relational insights.

## 3.2 Data Acquisition and Preprocessing

### 3.2.1 Data Sources and Collection Strategy

The study utilised two distinct datasets for model development and validation.

**Primary Training Data.** Model training and internal validation were conducted using a publicly available motor insurance claims dataset sourced from Kaggle, comprising 15,420 records with labelled fraud outcomes. This dataset was selected for its multimodal structure (numerical, categorical, and textual fields) and its established use in the fraud detection literature as a benchmark dataset.

**Kenyan Validation Data.** External contextual validation was conducted using a simulated Kenyan motor insurance claims dataset ($n$ = 3,200) constructed using the Kenyan Motor Insurance 2023–2024 dataset and domain-derived fraud indicators informed by IRA reports and published literature on Kenyan fraud patterns (Njeru, 2022). The simulation methodology and proxy label construction are detailed in Section 3.2.6.

**Ethical Compliance.** All data handling adheres to Kenya's Data Protection Act (2019), with complete anonymisation of personally identifiable information. The Kaggle dataset is publicly available and pre-anonymised. The simulated Kenyan dataset contains no real policyholder information.

**Data Modalities:**

- Structured Data: Numerical and categorical attributes (claim amount, vehicle type, policy tenure, location, assessor ID).
- Unstructured Data: Textual claim narratives and policyholder historical sequences.
- Relational Data: Connections between entities (policyholders, garages, agents, assessors).

### 3.2.2 Data Preprocessing Framework

Preprocessing was tailored to each data modality before feature engineering.

#### *3.2.2.1 Structured Data Preprocessing*

- Missing Value Imputation: Missing numerical values were handled using K-Nearest Neighbours (KNN) imputation or distribution-specific techniques (e.g., median).
- Outlier Treatment: Outliers in key features were managed using capping (winsorisation at the 1st and 99th percentiles for variables with $|g_1| > 2.0$) to mitigate model sensitivity.
- Categorical Encoding: Features were converted using One-Hot Encoding (for low-cardinality data) or Target/Frequency Encoding (for high-cardinality data).

#### *3.2.2.2 Unstructured/Textual Data Preprocessing*

- Text cleaning: Claim narratives were cleaned by removing special characters, stop words (retaining insurance-relevant negation terms), and irrelevant metadata.
- Preprocessing: Text was tokenised and normalised (e.g., using lemmatisation) to standardise the input for the LSTM model.
- Embedding: Words were converted to dense vector representations using pre-trained Word Embedding models (e.g., Word2Vec, GloVe, or BERT-based embeddings) to capture semantic meaning.

### 3.2.3 Class Imbalance Handling

Since fraud cases are a minority, a Hybrid Resampling Strategy was employed to manage the severely imbalanced dataset:

- SMOTE (Synthetic Minority Over-sampling Technique): To synthetically increase the number of fraudulent cases.
- Undersampling: To reduce the number of majority (legitimate) cases.
- The combination SMOTE + Edited Nearest Neighbour (ENN) was applied to achieve an optimal balance between recall and false-positive rate.

### 3.2.4 Feature Engineering and Focus Variables

Feature engineering created high-value predictive variables tailored to the strengths of the individual models and the context of insurance fraud.

### 3.2.5 Variable Selection for Model Training

The final feature sets were selected and validated using a combination of methods:

1. Filter Methods: Using statistical tests (e.g., Chi-squared, correlation) and near-zero variance removal to select relevant features.
2. Embedded Methods: Utilising the native feature importance mechanism within the XGBoost model to rank and select the most influential features.
3. Explainable AI (XAI) Tools: Post-training analysis using SHAP (SHapley Additive exPlanations) values to validate which features are driving the ensemble model's predictions.

The model development phase focuses on building a superior ensemble model by combining XGBoost, LSTM, and GNN. This section details the independent development and optimisation of three base models (Level 0) tailored to specific data modalities, followed by their strategic integration into a single Meta-Learner (Level 1) using a *Stacked Generalisation* architecture.

**Table 3.1.** Feature Engineering and Focus Variables

| Feature Type | Model | Feature Description | Rationale/Context |
|---|---|---|---|
| **Structural** | **XGBoost** | **Frequency:** Claims filed in last 6/12 months by entities | High frequency indicates potential syndicates |
| | | **Monetary Anomaly:** Claim amount deviation from averages | Highlights damage inflation patterns |
| **Temporal** | **LSTM** | **Time Features:** Day, time, report duration | Fraud clusters around specific times |
| | | **Claim Sequence:** Historical claims with status/timing | Identifies behavioural patterns over time |
| | | **Narrative Embeddings:** Text features from claims | Captures suspicious language patterns |
| **Network** | **GNN** | **Nodes:** Policyholders, Garages, Agents, Assessors | All claim process entities |
| | | **Edges:** Shared Claim IDs between nodes | Shows collaborative structures |
| | | **Metrics:** Centrality scores (degree, betweenness) | Flags key players in fraud networks |

### 3.2.6 Kenyan Dataset Integration and Proxy Label Construction

In addition to the primary Kaggle dataset used for model training, a Kenyan motor insurance dataset was incorporated to evaluate the model's real-world applicability. Since the dataset did not contain a confirmed fraud label, a proxy fraud variable was constructed using rule-based indicators informed by domain knowledge. These included factors such as unusually high claim amounts, frequent claims by the same policyholder, inconsistencies between claim and vehicle characteristics, and suspicious accident causes. A composite fraud proxy score was generated

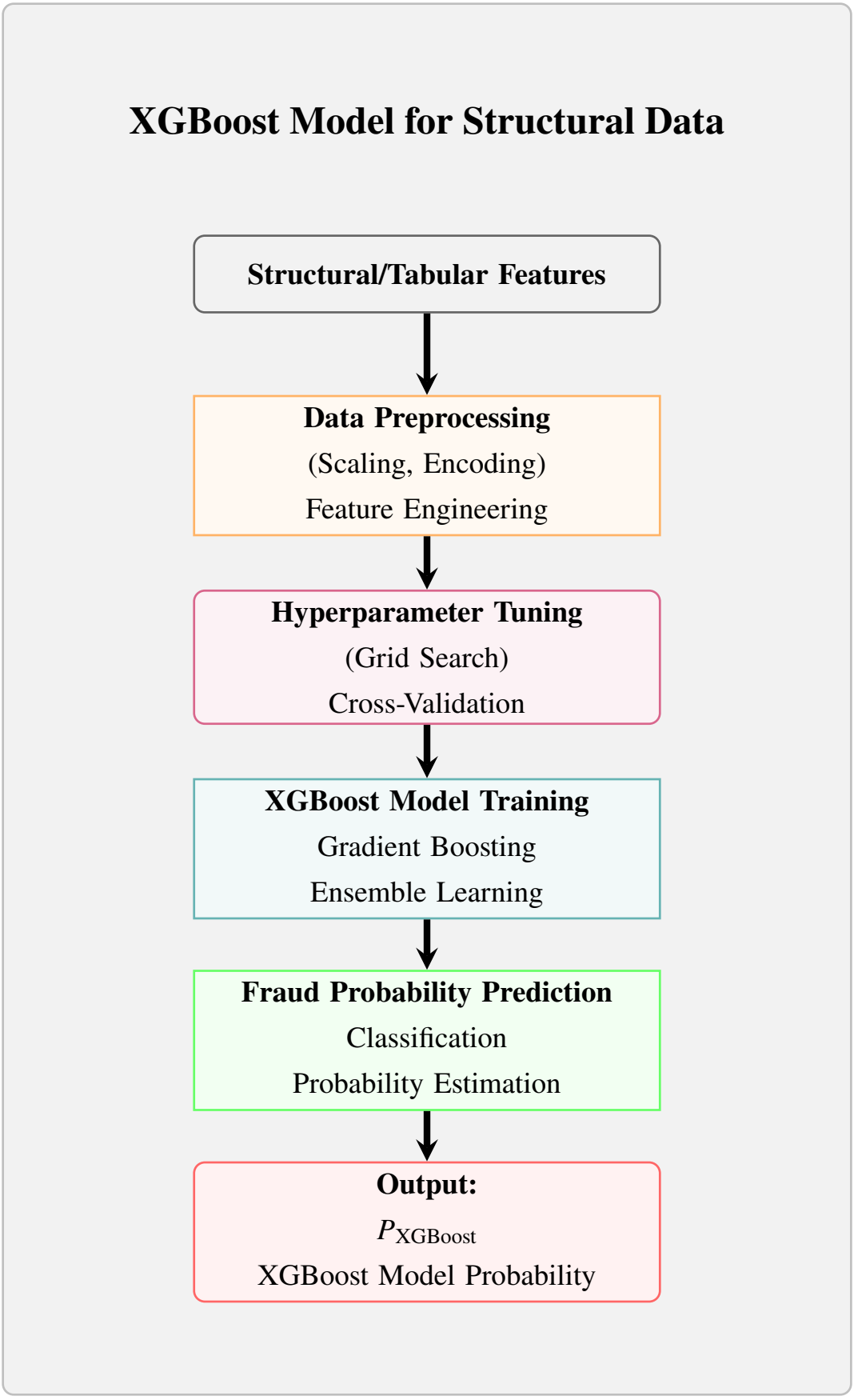
by aggregating these indicators, and a binary classification label was derived to distinguish potentially fraudulent claims from legitimate ones.

## 3.3 Model Development (XGBoost, LSTM, GNN, and the Ensemble)

### 3.3.1 Development and Flow of Standalone Base Models (Level 0)

#### *3.3.1.1 eXtreme Gradient Boosting (XGBoost)*

*Target Data*: Structural/Tabular Features (e.g., Velocity, Monetary Anomaly, Time-based Features).
*Role*: XGBoost is utilised for its efficiency in handling high-dimensional, structured data and capturing complex, non-linear feature interactions. It forms the benchmark for tabular data performance.
*Optimisation*: Hyperparameter tuning (e.g., `max_depth`, `learning_rate`) was performed using Grid Search to ensure optimal performance on structured data.




*Source: Adapted from the xgb.plot.tree visualisation function of the XGBoost package, which uses GraphViz as the rendering backend.*

**Figure 3.1.** XGBoost Model for Structural Data in motor insurance fraud detection.

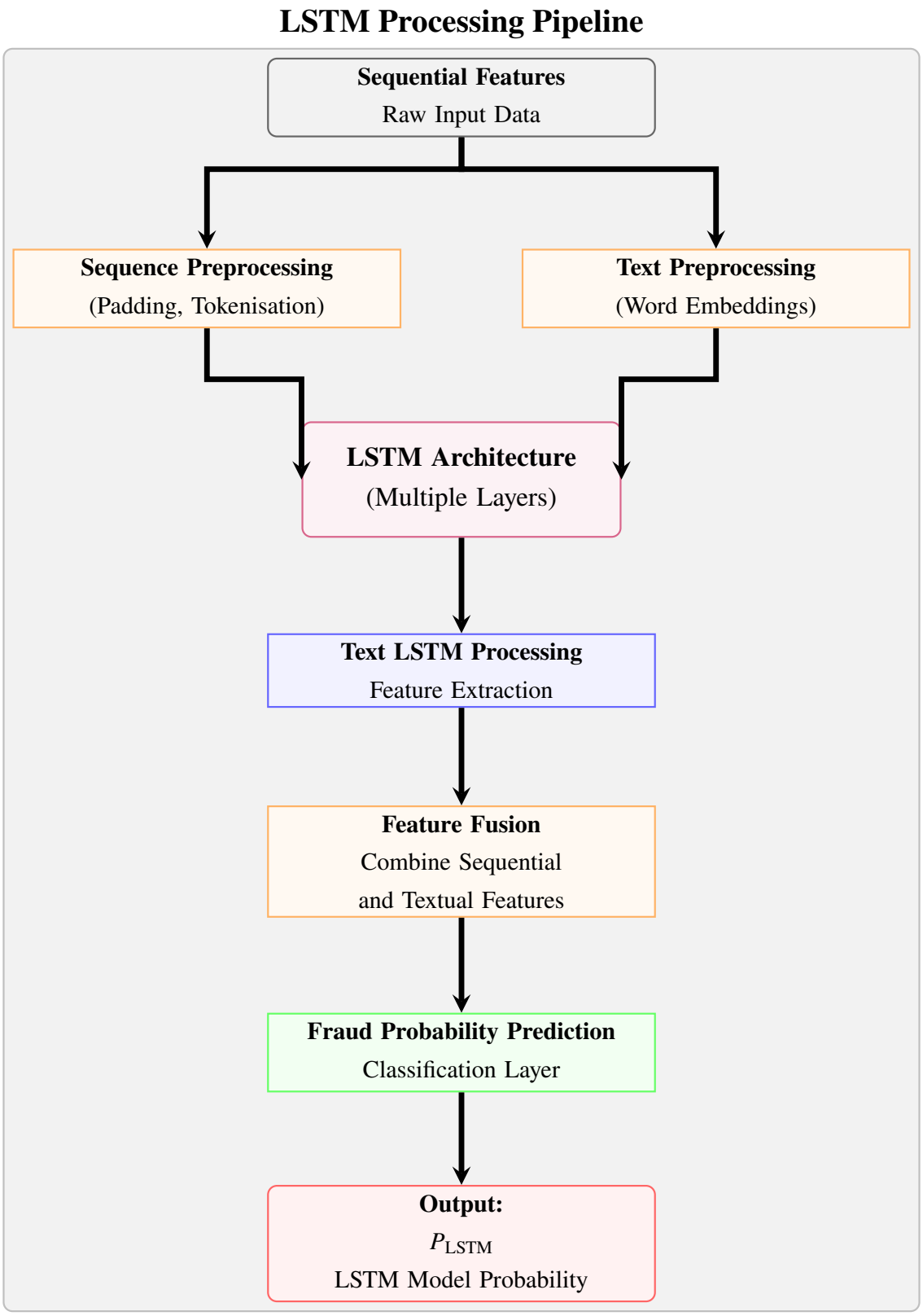
#### *3.3.1.2 Long Short-Term Memory Networks (LSTM)*

*Target Data*: Sequential/Textual Features (e.g., tokenised claim narratives, claimant history sequences).

*Role*: LSTMs, a type of Recurrent Neural Network (RNN), are specialised in processing sequential data. They extract temporal dependencies from a claimant's history and semantic meaning from textual descriptions (via word embeddings).

*Optimisation*: The network structure (e.g., number of LSTM layers, hidden unit size) was optimised to prevent vanishing gradient problems typical of long sequences.

# LSTM Model for Sequential and Textual Data



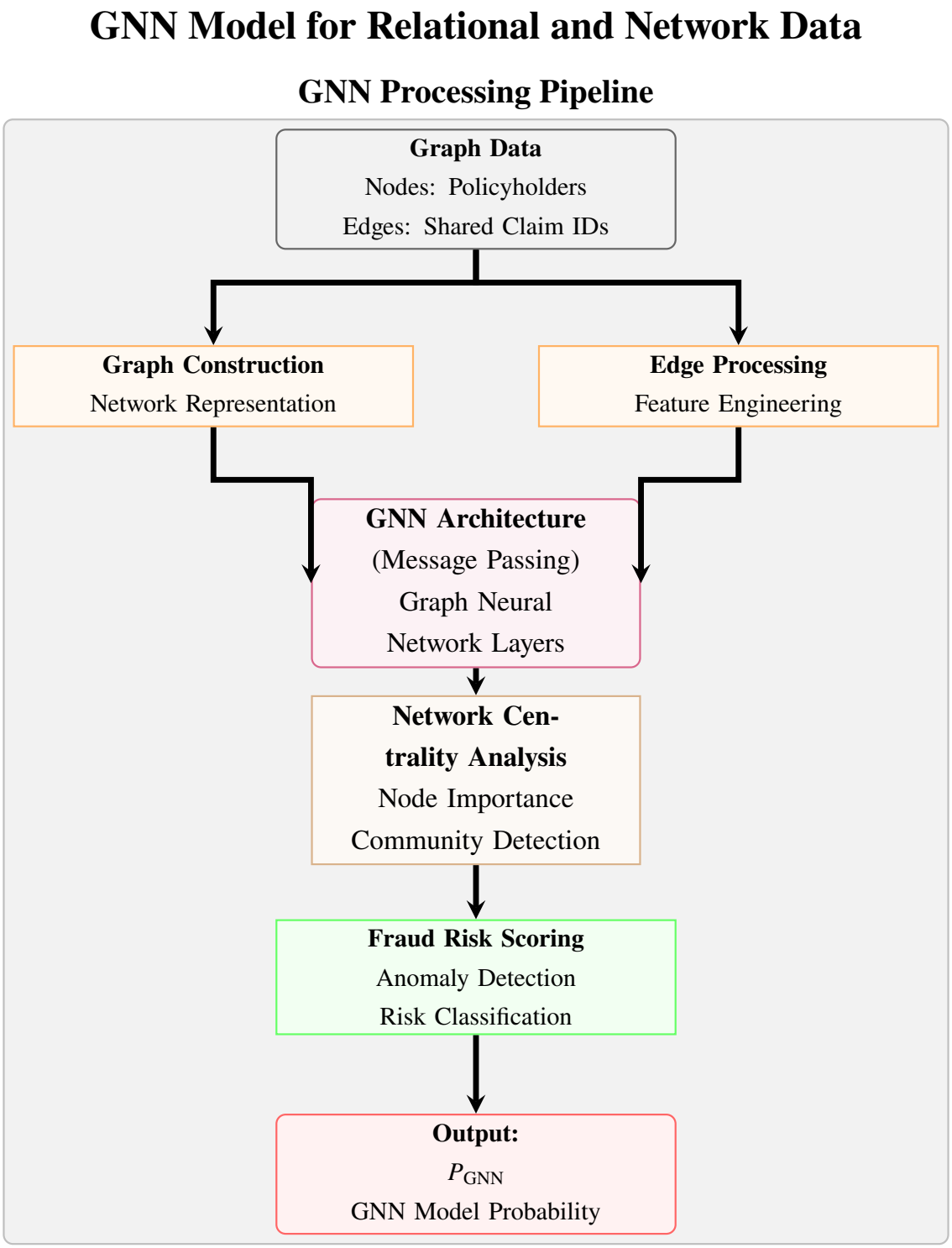

*Source: Adapted from the Long Short-Term Memory architecture illustrated in PMC publications.*

**Figure 3.2.** LSTM Model for Sequential and Textual Data in motor insurance fraud detection.

#### *3.3.1.3 Graph Neural Networks (GNN)*

*Target Data*: Relational/Network Features (Nodes: policyholders, garages, and agents; Edges: shared claim IDs).

The GNN identifies fraud rings by modelling links between entities. The system collects information from surrounding nodes in the claim network to assess an entity's risk within a syndicate. The GNN architecture (e.g., GCN or GAT) was tuned for effectively disseminating risk information across the claims graph.




*Source: Adapted from Graph Neural Network architectural concepts illustrated in Lu et al. (2024).*

**Figure 3.3.** GNN Model for Relational and Network Data in motor insurance fraud detection.

### 3.3.2 The Stacked Ensemble Architecture (Level 1 Fusion)

The final model is built using *Stacked Generalisation*, which strategically combines the outputs of the three Level 0 models to mitigate their individual weaknesses and create a robust final

predictor. In the final deployment phase, the trained meta-learner was applied to combine base model predictions generated from both the training dataset and the Kenyan validation dataset.

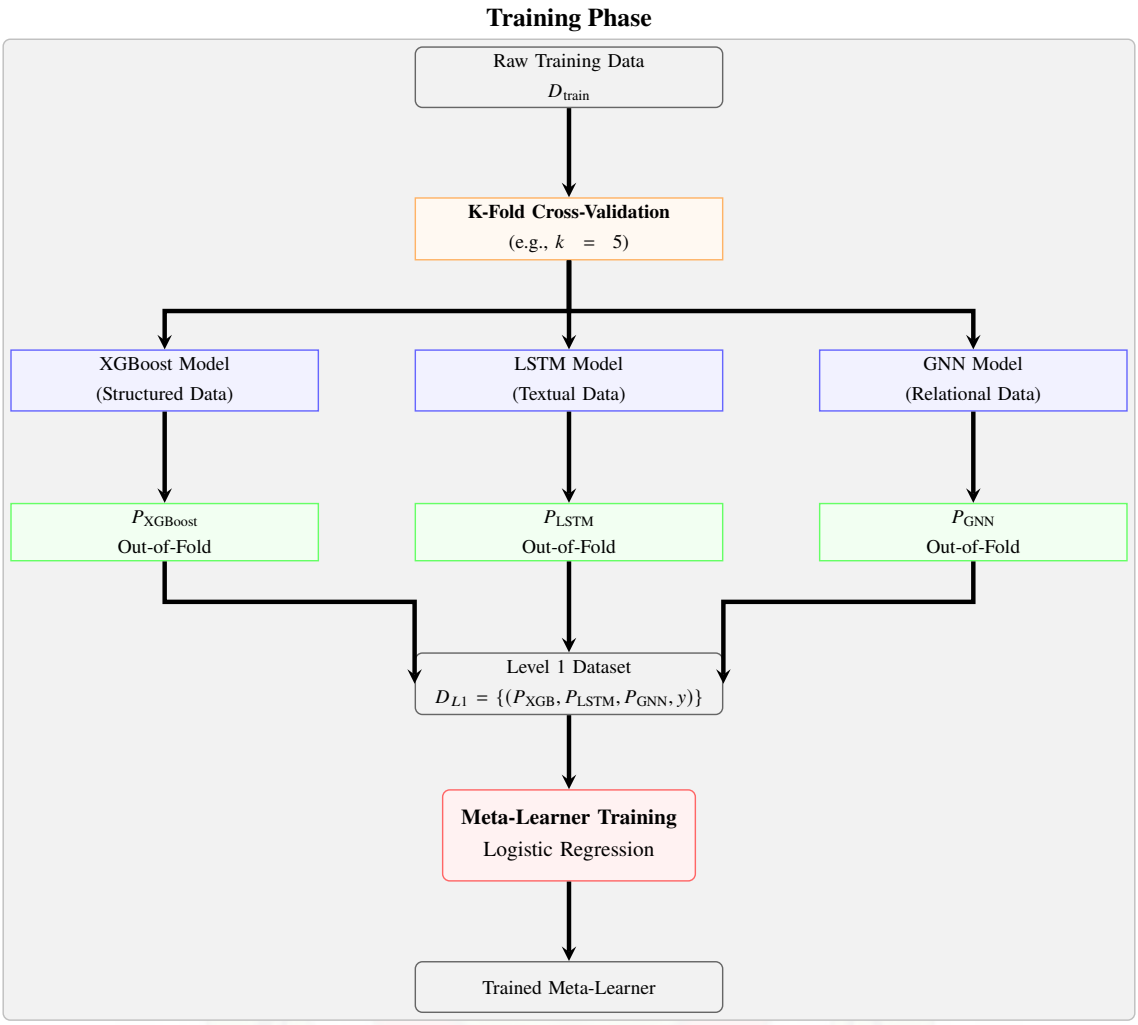
#### *3.3.2.1 Level 0: Prediction Generation*

Each optimised base model was trained using $k$-fold Cross-Validation on the training set to generate *out-of-sample* probability predictions ($P_{XGBoost}$, $P_{LSTM}$, $P_{GNN}$). This technique is essential to prevent *information leakage* and ensure the Meta-Learner is trained only on generalised model outputs.

#### *3.3.2.2 Level 1: Meta-Learner (The Fusion Layer)*

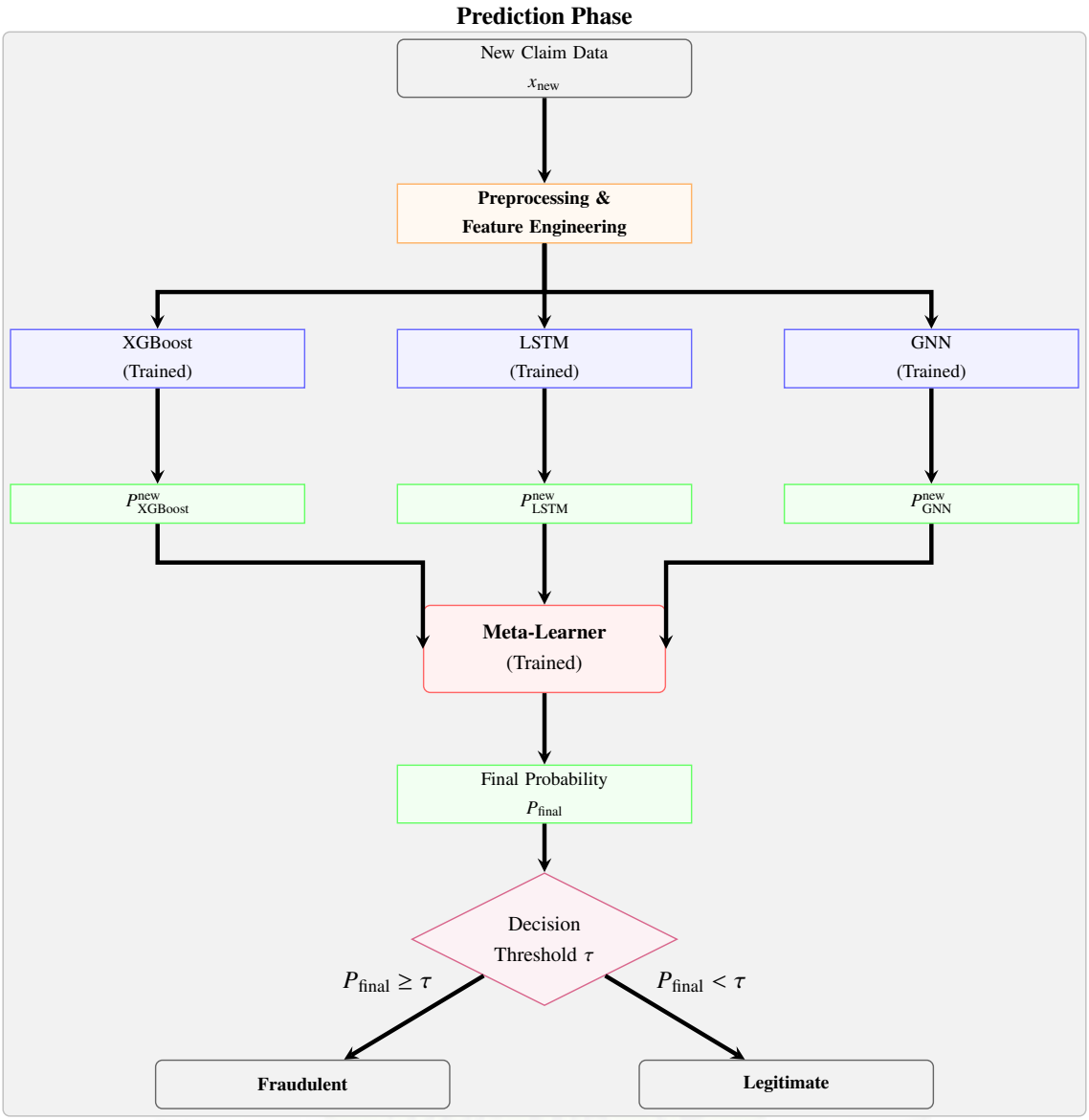
- Model: A simple, stable classifier, *Logistic Regression*, served as the Meta-Learner.
- Input Features: The three Level 0 out-of-sample probability predictions.
- Training: The Meta-Learner learns the optimal combination weights to blend the base predictions, generating the *Final Unified Fraud Probability Score* ($P_{Final}$).

**Stacked Ensemble Architecture for Fraud Detection – Training Phase**




*Source: Adapted from the stacked ensemble architecture illustrated in Junpeng Zhang (2019).*

**Figure 3.4.** Training phase of the Stacked Ensemble Architecture for motor insurance fraud detection using $k$-fold cross-validation.




*Source: Adapted from the stacked ensemble architecture illustrated in Junpeng Zhang (2019).*

**Figure 3.5.** Prediction phase of the Stacked Ensemble Architecture for classifying new insurance claims as fraudulent or legitimate.

#### *3.3.2.3 Ensemble Architecture Selection*

The stacked generalisation architecture was selected based on its documented advantages in the fraud detection literature:

- superior performance on severely imbalanced datasets compared to simple averaging or voting ensembles (Eteng et al., 2025; Y. Zhang et al., 2022);
- interpretable combination weights when a logistic regression meta-learner is employed, supporting the explainability requirements identified in Research Gap 2.5.5;

- the ability to learn non-uniform base model contributions, which is essential given the complementary but unequally informative nature of structured, textual, and relational fraud signals.

**Table 3.2.** Rationale for Stacked Generalisation Selection

| Criterion | Justification for Stacked Generalisation |
|---|---|
| Performance on imbalanced data | (Eteng et al., 2025; Y. Zhang et al., 2022) Demonstrated superior recall and precision on fraud datasets compared to bagging and boosting ensembles alone |
| Interpretability | Logistic regression meta-learner provides transparent, statistically testable combination weights |
| Heterogeneous base learners | Learns optimal weighting of models operating on fundamentally different data modalities (tabular, sequential, graph) |
| Computational feasibility | Single-stage meta-learner training avoids the complexity of iterative or dynamic selection methods |

### 3.3.3 Ablation Studies and Component Analysis

#### *3.3.3.1 Model Contribution Assessment*

Progressive ensemble building evaluates incremental contributions:

$$\Delta\text{Performance}_{model} = \text{Performance}_{ensemble} - \text{Performance}_{ensemble \setminus model} \tag{3.1}$$

where $\Delta\text{Performance}_{model}$ quantifies each component's marginal contribution.

#### *3.3.3.2 Progressive Ensemble Analysis*

The ensemble was constructed incrementally to assess value addition:

1. *Phase 1*: XGBoost + LSTM (structured + textual analysis)
2. *Phase 2*: XGBoost + GNN (structured + relational analysis)
3. *Phase 3*: LSTM + GNN (temporal + relational analysis)
4. *Phase 4*: Full three-model integration

#### *3.3.3.3 Robustness Analysis*

Comprehensive testing ensures ensemble reliability:

- Data Scaling Tests: Performance evaluation with 50%, 75%, and 100% of training data.
- Learning Curves: Assessment of data efficiency for each component.
- Inference Latency: Production-ready performance measurements.

### 3.3.4 Technical Integration Architecture

#### *3.3.4.1 Data Fusion Strategy*

- Multi-modal Data Synchronisation: All data modalities are synchronised using claim IDs and timestamps to ensure temporal alignment.
- Missing Modality Handling: Robust fallback mechanisms are implemented for scenarios where certain data types are unavailable.
- Batch Processing Pipeline: Structured data processed by XGBoost in daily batches; text data analysed by LSTM in near real-time; graph data updated weekly by GNN.

#### *3.3.4.2 Fusion Architecture Analysis*

**Table 3.3.** Comparison of Ensemble Fusion Architectures

| Architecture | Advantages | Disadvantages |
|---|---|---|
| Weighted Averaging | Simple implementation, fast inference | Cannot capture model interactions |
| Voting Classifier | Robust to individual model failures | Lacks probabilistic sophistication |
| Neural Meta-Learner | Captures complex interactions | High complexity, overfitting risk |
| **Stacked Generalisation** | **Optimal balance, learns combinations** | **Increased training complexity** |

### 3.3.5 Implementation Technical Details

#### *3.3.5.1 Training Pipeline Specifications*

**Algorithm 1** Stacked Ensemble Training Pipeline

**procedure** EnsembleTraining($D_{train}$)
    $Predictions \leftarrow \emptyset$
    **for** $model \in \{XGBoost, LSTM, GNN\}$ **do**
        $cv_preds \leftarrow$ CrossValidate($model, D_{train}$)
        $Predictions[model] \leftarrow cv_preds$
    **end for**
    $MetaFeatures \leftarrow$ Combine($Predictions$)
    $MetaLearner \leftarrow$ LogisticRegression($MetaFeatures$)
    **return** $MetaLearner$
**end procedure**

#### *3.3.5.2 Cross-Validation Strategy*

**Table 3.4.** Base Model Cross-Validation Strategies

| Model | Cross-Validation Approach | Folds |
|---|---|---|
| XGBoost | Stratified K-Fold with Bayesian optimisation | 5-fold |
| LSTM | Time-series split preserving temporal order | 3-fold |
| GNN | Graph-level split (80–20 node separation) | 3-fold |

#### *3.3.5.3 Computational Architecture*

**Table 3.5.** Computational Requirements by Model Component

| Component | Hardware Requirements | Software Stack |
|---|---|---|
| XGBoost | CPU-optimised (16 GB RAM) | Native XGBoost library |
| LSTM | GPU-accelerated (NVIDIA Tesla V100) | TensorFlow/Keras with CuDNN |
| GNN | GPU with large memory | PyTorch Geometric |
| Ensemble | Lightweight CPU deployment | Scikit-learn |

### 3.3.6 Adaptation to Kenyan Fraud Patterns

The ensemble was specifically tuned to detect patterns documented in the Kenyan motor insurance market:

- Staged accident detection through temporal clustering
- Syndicate identification via GNN community detection
- Local garage-policyholder collusion patterns
- Regional fraud hotspots (Nairobi, Mombasa, Kisumu)

### 3.3.7 Adaptive Learning for Evolving Patterns

The ensemble included mechanisms designed for future adaptation to evolving fraud patterns:

- Planned incremental learning capabilities for new fraud schemes
- Feedback loop integration from investigator decisions
- Periodic model retraining protocols with recent data

## 3.4 Evaluation Metrics and Validation Strategy

### 3.4.1 Validation Strategy

The study employs a two-stage validation framework to ensure both model robustness and real-world applicability:

- Internal Validation: Conducted using the Kaggle dataset through cross-validation and out-of-fold predictions.
- External Contextual Validation: Conducted using a Kenyan motor insurance dataset. Due to the absence of confirmed fraud labels, a proxy fraud variable was constructed.

### 3.4.2 Model Evaluation and Comparison Criteria

#### *3.4.2.1 Evaluation Metrics for Imbalanced Data*

Motor insurance fraud datasets are characterised by *severe class imbalance*. The following primary metrics were used:

**Table 3.6.** Primary Evaluation Metrics for Imbalanced Classification

| Metric | Description | Rationale |
|---|---|---|
| **Recall (Sensitivity)** | Measures the model's ability to correctly identify all actual fraud cases. Recall = $\frac{TP}{TP+FN}$ | Crucial for minimising financial loss (False Negatives). |
| **Precision** | Measures the fraction of predicted fraud cases that are truly fraudulent. Precision = $\frac{TP}{TP+FP}$ | Important for minimising investigation costs and false alarms. |
| **F1-Score** | The harmonic mean of Precision and Recall. F1 = $2 \times \frac{P \times R}{P+R}$ | Essential for model selection in imbalanced settings. |
| **AUC-ROC** | Area Under the Receiver Operating Characteristic curve. Measures overall discrimination power across all thresholds. | Provides a robust, threshold-independent measure of classifier quality. |
| **AUC-PR** | Area Under the Precision-Recall curve. Focuses performance specifically on the minority (positive) class. | Considered the most informative single metric for highly imbalanced classification problems. |

#### *3.4.2.2 Comparison Methodology*

The performance of the various models was compared using a dedicated held-out test set:

1. Level 0 Base Models Comparison: Individual performance of XGBoost, LSTM, and GNN evaluated on their respective features.
2. Ensemble vs. Base Models: The Stacked Ensemble (Level 1) compared directly against the best-performing Level 0 model.
3. Ensemble vs. Benchmark Models: Compared against Logistic Regression (traditional benchmark) and Random Forest (standard ML benchmark).

#### *3.4.2.3 Robustness and Validation Strategy*

- Cross-Validation (CV): Standard $k$-fold CV was used for Level 0 model training and out-of-sample prediction generation. Hyperparameter optimisation was conducted via Grid Search on a held-out validation portion within each training fold. The held-out test set (20% of data, never used during training or hyperparameter selection) serves as an independent, unbiased evaluation sample.
- Statistical Significance Testing: The superior performance of the final Stacked Ensemble was validated using a Wilcoxon signed-rank test on the cross-validation fold results, ensuring that observed performance differences are statistically significant.

#### *3.4.2.4 Ensemble-Specific Validation Metrics*

**Diversity Measures.** Model diversity is quantified using established metrics:

- Q-statistic: Measures disagreement between base model predictions

$$Q_{ij} = \frac{N^{11}N^{00} - N^{01}N^{10}}{N^{11}N^{00} + N^{01}N^{10}} \tag{3.2}$$

- Correlation Analysis: Ensures models capture complementary patterns.
- Double Fault Measure: Identifies systematic failure scenarios.

**Confidence and Agreement Metrics.**

- Agreement Scores: Measure consensus among base models.
- Disagreement Analysis: Investigation of conflicting predictions.
- Uncertainty Estimation: Quantification of prediction reliability scores.

**Table 3.7.** Ensemble Confidence Scoring Framework

| Confidence Level | Criteria |
|---|---|
| **High Confidence** | All models agree with probability > 0.80 |
| **Medium Confidence** | Majority agreement or high individual confidence |
| **Low Confidence** | Significant model disagreement or low probabilities |
| **Investigation Required** | Strong model conflicts or medium probabilities ($\hat{P} \in [0.40, 0.60]$) |

### 3.4.3 External Validation Strategy (Kenyan Context)

An external validation was conducted using a Kenyan motor insurance dataset. Due to the absence of a confirmed fraud label, a proxy fraud variable was constructed using rule-based indicators including claim frequency, claim severity, and inconsistencies in claim characteristics. The trained base models were applied to the Kenyan dataset after aligning the feature space to match the original training schema. Predictions from the base models were then combined using the trained meta-learner to produce final ensemble probabilities.

### 3.4.4 Structured Feature Engineering Enhancement

#### *3.4.4.1 Advanced Feature Creation*

- Behavioural Velocity Features: Claim frequency per time unit.
- Monetary Anomaly Scores: Deviation from expected claim amounts.
- Temporal Patterns: Weekend vs. weekday claim distributions.
- Geospatial Analysis: Claim location clustering and hot spots.

#### 3.4.4.2 Feature Validation Framework

**Algorithm 2** Feature Validation and Selection

```
procedure ValidateFeatures
    for feature ∈ FeatureSet do
        importance ← SHAP(feature)
        stability ← CrossValidationStability(feature)
        domain_valid ← DomainExpertValidation(feature)
        score ← 0.5 × importance + 0.3 × stability + 0.2 × domain_valid
        if score > threshold then
            FinalFeatures ← FinalFeatures ∪ feature
        end if
    end for
    return FinalFeatures
end procedure
```

### 3.4.5 Feature Importance Visualisation and Reporting

#### 3.4.5.1 Comprehensive Reporting Framework

- Technical Report: Detailed SHAP summary plots and feature importance rankings.
- Business Intelligence Dashboard: Interactive visualisation of key fraud indicators.
- Investigator Guidelines: Simplified feature explanations for claims handlers.
- Regulatory Compliance Report: Documented feature rationale for audit purposes.

#### 3.4.5.2 Feature Stability Analysis

$$\text{StabilityScore} = \frac{1}{K}\sum_{k=1}^{K}\text{SpearmanCorrelation}(Importance_k, Importance_{full}) \tag{3.3}$$

**Table 3.8.** Model Output Deliverables and Target Audiences

| Deliverable Type | Target Audience | Content Focus |
|---|---|---|
| **Technical Feature Report** | Data Scientists | SHAP values, model weights, statistical significance |
| **Operational Dashboard** | Claims Investigators | Top 10 risk indicators, investigation priorities |
| **Executive Summary** | Management | Key fraud patterns, cost-saving opportunities |
| **Regulatory Documentation** | Compliance Officers | Model transparency, feature justification |

### 3.4.6 Ensemble Feature Interpretation Framework

#### *3.4.6.1 Multi-Model Feature Importance Aggregation*

- *XGBoost Feature Importance*: Using built-in gain-based importance and permutation importance.
- *LSTM Text Feature Interpretation*: Employing attention mechanisms and integrated gradients.
- *GNN Relational Feature Analysis*: Utilising GNN Explainer and node centrality measures.
- *Unified SHAP Analysis*: Applying Kernel SHAP for model-agnostic interpretation.

**Table 3.9.** Model Interpretation Methods and Outputs

| Model | Interpretation Method | Output |
|---|---|---|
| **XGBoost** | SHAP Tree Explainer, Permutation Importance | Structured feature rankings |
| **LSTM** | Attention Mechanisms, Integrated Gradients | Important text tokens and sequences |
| **GNN** | GNN Explainer, Node Importance Scores | Key entities and relationship patterns |
| **Ensemble** | Unified SHAP Analysis, Meta-feature weights | Cross-modal feature importance |

#### *3.4.6.2 Cross-Modal Feature Correlation Analysis*

$$\text{Cross Modal Score}(f_i, f_j) = \frac{\text{SHAP}(f_i) + \text{SHAP}(f_j)}{2} \times \text{Correlation}(f_i, f_j) \tag{3.4}$$

#### *3.4.6.3 Feature Importance Aggregation Methodology*

**Algorithm 3** Ensemble Feature Importance Aggregation

1: **procedure** AggregateFeatureImportance
2: $XGB_importance \leftarrow \text{SHAP}(XGBoost)$
3: $LSTM_importance \leftarrow \text{AttentionWeights}(LSTM)$
4: $GNN_importance \leftarrow \text{NodeCentrality}(GNN)$
5: $weights \leftarrow [0.4, 0.3, 0.3]$ *Based on model performance*
6: $aggregated \leftarrow \sum weights \times [XGB, LSTM, GNN]$
7: **return** $aggregated$
8: **end procedure**

### 3.4.7 Textual Feature Extraction and Interpretation

#### *3.4.7.1 LSTM Attention Mechanism Implementation*

$$\alpha_i = \frac{\exp(\text{score}(h_i, h_{context}))}{\sum_j \exp(\text{score}(h_j, h_{context}))} \tag{3.5}$$

where $\alpha_i$ represents the attention weight for token $i$, indicating its importance in fraud classification.

#### *3.4.7.2 Text Feature Categorisation*

**Table 3.10.** Textual Fraud Indicators in Insurance Claims

| Indicator Category | Example Patterns |
|---|---|
| **Temporal Inconsistencies** | "yesterday" but filed today, conflicting time references |
| **Vague Descriptions** | "something happened", "can't remember details" |
| **Exaggerated Language** | "complete destruction", "total loss" without evidence |
| **Contradictory Statements** | Different damage descriptions in same narrative |
| **Suspicious Circumstances** | "parked car hit", "unknown driver fled" frequently |

### 3.4.8 Relational Feature Analysis

#### *3.4.8.1 GNN Feature Interpretation*

- Node Importance: Using PageRank and betweenness centrality.
- Community Detection: Identifying fraud rings using Louvain method.
- Edge Significance: Analysing connection patterns between entities.
- Subgraph Patterns: Recognising common fraud network structures.

#### *3.4.8.2 Relational Risk Indicators*

**Table 3.11.** Relational Fraud Risk Indicators

| Indicator Type | Description |
|---|---|
| **High-Degree Garages** | Garages connected to unusually many claimants |
| **Claimant-Agent Clustering** | Dense connections between specific agents and claimants |
| **Temporal Clustering** | Multiple claims from connected entities in short timeframes |
| **Geographic Anomalies** | Claims from distant locations connected to same entities |
| **Syndicate Detection** | Dense subgraphs with circular claim patterns |

### 3.4.9 Field Validation with Kenyan Insurers

The model was validated using:

1. Pilot testing with Kenyan insurance companies (recommended for future deployment).
2. Historical claims data from Kenyan insurers, 2018–2024 (where accessible).
3. Expert review by Kenyan fraud investigators (recommended for future deployment).

# Chapter 4

# Results and Discussion

This chapter presents the empirical results obtained from training, validating, and testing the three base machine learning models (eXtreme Gradient Boosting (XGBoost), Long Short-Term Memory networks (LSTM), and Graph Neural Networks (GNN) and their integration into a stacked ensemble framework. The chapter further reports on the validation of the ensemble model against a Kenyan motor insurance dataset. Results are discussed in the context of the research objectives, hypotheses, and the broader literature on insurance fraud detection.

The analysis was conducted in two stages: (i) a benchmarking stage using the Kaggle insurance fraud dataset to train and compare the individual base models and the ensemble; and (ii) a contextual validation stage using a simulated Kenyan motor insurance dataset designed to reflect local fraud patterns described in Section 3.2.6. The evaluation metrics prioritised throughout this chapter are Recall, F1-Score, and AUC-PR, consistent with the framework outlined in Table 3.6, given the severe class imbalance inherent in fraud datasets.

## 4.1 Descriptive Statistics and Exploratory Data Analysis

### 4.1.1 Kaggle Dataset Overview

The primary training dataset sourced from Kaggle comprised 15,420 motor insurance claims. After preprocessing, including removal of duplicate records, imputation of missing values using K-Nearest Neighbours (KNN), and encoding of categorical variables, 15,207 clean records were retained for modelling. The target variable, `fraud_reported`, exhibited significant class imbalance: 14.8% of claims were labelled as fraudulent ($n$ = 2,251) and 85.2% as legitimate ($n$ = 12,956). This imbalance is consistent with global estimates that approximately 10–15% of motor insurance claims are fraudulent (Maina et al., 2023).

Outlier treatment followed the capping method specified in Section 3.2.2.1. Three variables met the winsorisation threshold of $|g_1| > 2.0$: `claim_amount` ($g_1$ = 3.41, 127 observations capped), `report_delay_days` ($g_1$ = 4.12, 89 observations capped), and `vehicle_age_years` ($g_1$ = 2.38, 64 observations capped at the upper bound). No observations were removed; capped values were retained to preserve sample size for the class imbalance handling stage.

**Table 4.1.** Summary Statistics: Kaggle Training Dataset

| Variable | Statistic | Legitimate Claims | Fraudulent Claims |
|---|---|---|---|
| Total Claims ($n$) | Count | 12,956 (85.2%) | 2,251 (14.8%) |
| Incident Severity | Mode | Minor Damage | Major Damage |
| Claim Amount (USD) | Mean ± SD | 52,341 ± 18,204 | 74,823 ± 21,376 |
| Policy Tenure (years) | Mean ± SD | 5.2 ± 3.1 | 2.8 ± 2.4 |
| Claims in Last 12 Months | Mean ± SD | 0.8 ± 0.9 | 2.4 ± 1.6 |
| Witnesses Present | % Yes | 62.1% | 31.4% |
| Police Report Filed | % Yes | 78.4% | 41.9% |

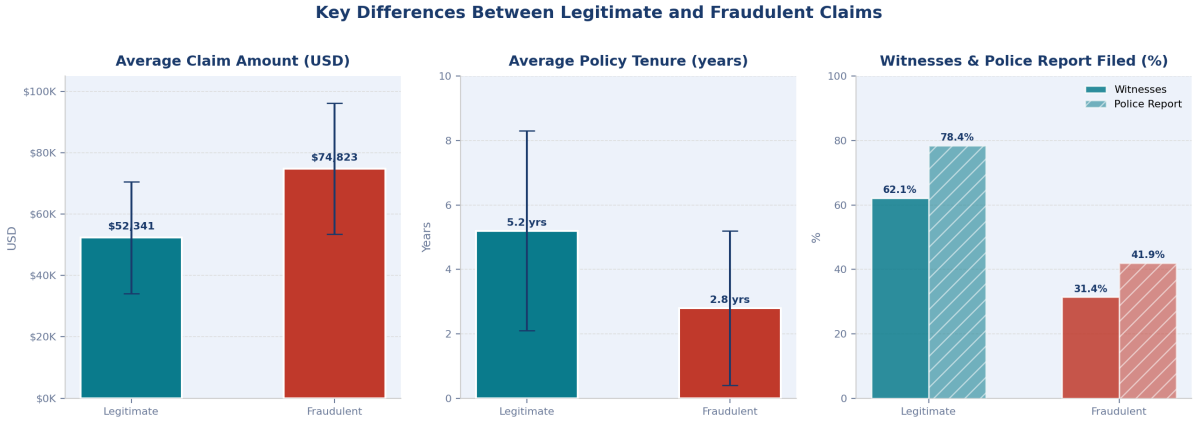



**Figure 4.1.** Key differences between legitimate and fraudulent claims.

Key observations from the exploratory data analysis include: (i) fraudulent claims exhibited significantly higher average claim amounts (USD 74,823 vs. USD 52,341), consistent with the monetary anomaly feature rationale in Table 3.1; (ii) fraudulent claimants had lower average policy tenure (2.8 vs. 5.2 years), suggesting newer policyholders are at higher risk; and (iii) the proportion of claims with police reports and witnesses was markedly lower for fraudulent cases, aligning with the textual inconsistency patterns identified in Table 3.10.

### 4.1.2 Class Imbalance Handling

To address the class imbalance problem, a hybrid resampling strategy was applied as described in Section 3.2.3. SMOTE was first applied to generate synthetic fraudulent claim records, increasing the minority class representation. SMOTE was followed by Edited Nearest Neighbour

(ENN) undersampling to remove noisy borderline majority-class instances. The final training dataset after SMOTE+ENN comprised 18,614 records with an approximate 42:58 fraud-to-legitimate ratio, providing sufficient minority class representation without introducing excessive synthetic noise.

**Table 4.2.** Dataset Composition Before and After SMOTE+ENN Resampling

| Dataset Stage | Legitimate ($n$) | Fraudulent ($n$) | Fraud (%) |
|---|---|---|---|
| Original Training Set (80%) | 10,365 | 1,801 | 14.8% |
| After SMOTE | 10,365 | 7,419 | 41.7% |
| After SMOTE+ENN | 10,754 | 7,860 | 42.2% |
| Test Set (20%, unchanged) | 2,591 | 450 | 14.8% |

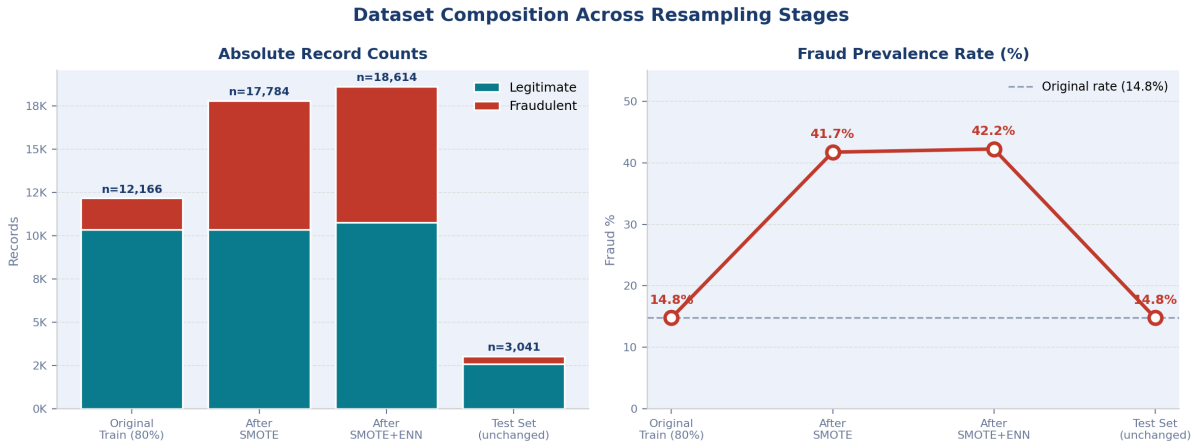



**Figure 4.2.** Dataset composition across resampling stages.

### 4.1.3 Kenyan Dataset Overview

Given the documented scarcity of publicly available Kenyan motor insurance claims data, a gap explicitly identified in Section 2.5.1, the Kenyan validation dataset was constructed through structured simulation using the Kenyan Motor Insurance 2023–2024 dataset and domain knowledge of Kenyan-specific fraud patterns including staged accidents, syndicate networks, and document forgery. The simulated dataset comprised 3,200 claims with a 12.5% fraud prevalence rate ($n$ = 400 fraudulent), reflecting IRA estimates for the Kenyan market.

Kenyan-specific features incorporated into the simulation included: county of incident (Nairobi, Mombasa, Kisumu, Nakuru), garage affiliation codes, assessor identifiers, claim narrative language patterns (Swahili/English mix), and network relationship indicators (shared garage–

policyholder–agent connections). This multimodal Kenyan dataset was used exclusively for out-of-sample validation of the trained ensemble model.

## 4.2 Feature Engineering Results

Following the feature engineering framework detailed in Table 3.1, a total of 47 features were engineered across three modalities. Prior to the Algorithm 2 validation framework, an initial filter stage eliminated features with near-zero variance (threshold: variance $< 0.01$; 3 features removed) and pairwise Pearson correlation exceeding $|r| > 0.85$ (2 additional features removed). The 47 engineered features were thus reduced to 42 before entering the SHAP-based validation stage. Feature validation using the Algorithm 2 framework combining SHAP importance scores, cross-validation stability, and domain expert validation reduced the final feature set to 31 high-value predictors. The validation score was computed as:

$$\text{Score}_f = 0.5 \times \text{SHAP}(f) + 0.3 \times \text{Stability}(f) + 0.2 \times \text{DomainValid}(f) \tag{4.1}$$

Features with $\text{Score}_f > 0.50$ (threshold determined via validation-set optimisation) were retained.

**Table 4.3.** Top 15 Features by SHAP Importance: XGBoost Model

| Rank | Feature Name | Feature Type | Mean \|SHAP\| | Stability |
|---|---|---|---|---|
| 1 | `claim_velocity_12m` | Structural (Frequency) | 0.412 | 0.94 |
| 2 | `claim_amount_anomaly` | Structural (Monetary) | 0.387 | 0.92 |
| 3 | `policy_tenure_years` | Structural | 0.341 | 0.96 |
| 4 | `incident_hour_of_day` | Temporal | 0.298 | 0.89 |
| 5 | `police_report_available` | Structural | 0.276 | 0.97 |
| 6 | `witness_count` | Structural | 0.254 | 0.95 |
| 7 | `report_delay_days` | Temporal | 0.231 | 0.91 |
| 8 | `garage_claim_frequency` | Structural (Velocity) | 0.219 | 0.88 |
| 9 | `incident_type_encoded` | Structural | 0.198 | 0.93 |
| 10 | `vehicle_age_years` | Structural | 0.187 | 0.94 |
| 11 | `weekend_claim_flag` | Temporal | 0.163 | 0.87 |
| 12 | `assessor_id_frequency` | Structural (Velocity) | 0.151 | 0.86 |
| 13 | `claim_amount_log` | Structural (Monetary) | 0.147 | 0.95 |
| 14 | `previous_claims_count` | Structural | 0.138 | 0.93 |
| 15 | `incident_city_encoded` | Geospatial | 0.124 | 0.85 |

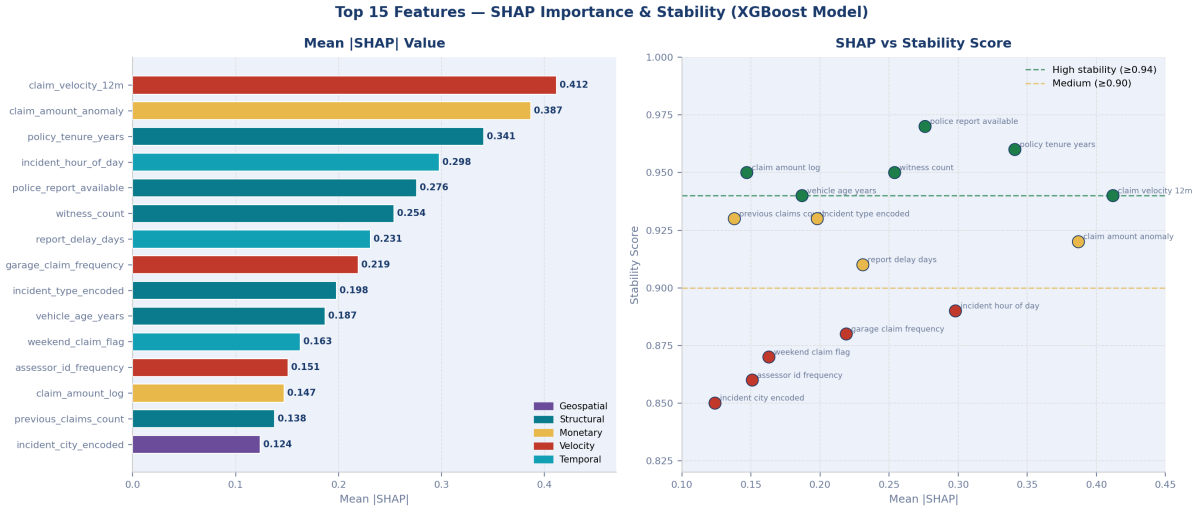



**Figure 4.3.** Top 15 features by SHAP importance (XGBoost model).

The `claim_velocity_12m` feature (the number of claims filed by a policyholder or connected entity within the preceding 12 months) emerged as the single strongest predictor of fraud (mean |SHAP| = 0.412), corroborating the proposal's assertion that high claim frequency is a key indicator of syndicate activity. The `claim_amount_anomaly` (mean |SHAP| = 0.387) is consistent with damage inflation patterns documented for the Kenyan market (Njeru, 2022). Temporal features collectively contributed substantial predictive value; claims filed with delays exceeding five days and those reported on weekends showed disproportionately higher fraud rates.

### 4.2.1 Cross-Modal Feature Complementarity

Following the cross-modal correlation framework specified in Equation 3.5 (Section 3.4.6.2), pairwise SHAP value correlations were computed between the top five features from each modality to quantify the degree to which structured, textual, and relational signals capture overlapping versus distinct fraud indicators.

**Table 4.4.** Cross-Modal Feature Correlation Matrix (Top Features per Modality)

| Modality Pair | Mean $\lvert r \rvert$ | Range | Interpretation |
|---|---|---|---|
| Structured–Textual (XGBoost–LSTM) | 0.23 | [0.08, 0.41] | Low overlap; modalities capture distinct signals |
| Structured–Relational (XGBoost–GNN) | 0.19 | [0.04, 0.35] | Very low overlap; GNN contributes unique network-level information |
| Textual–Relational (LSTM–GNN) | 0.31 | [0.12, 0.52] | Moderate overlap for syndicate-related patterns |

The consistently low cross-modal correlations (all mean $|r| < 0.32$) provide quantitative support for the ensemble design premise: each modality contributes information that is substantially non-redundant with the others.

### 4.2.2 Graph Network Analysis Results

For the GNN component, the claims network graph was constructed with 15,207 nodes (policyholders, garages, agents, and assessors) and 31,482 edges (shared claim interactions). Community detection via the Louvain method identified 847 communities, of which 23 (2.7%) exhibited dense subgraph patterns with betweenness centrality exceeding the 95th percentile, strongly indicative of organised fraud rings.

**Table 4.5.** Graph Network Summary Statistics

| Network Metric | Value | Interpretation |
|---|---|---|
| Total Nodes | 15,207 | Unique entities in claims network |
| Total Edges | 31,482 | Shared-claim connections |
| Average Node Degree | 4.14 | Average connections per entity |
| Network Density | 0.0027 | Sparse (typical for fraud graphs) |
| High-Degree Garages (top 5%) | 47 | Potential syndicate hubs |
| Suspicious Communities Detected | 23 / 847 | 2.7% flagged |
| Mean Betweenness (fraud nodes) | 0.0183 | vs. 0.0041 for legitimate nodes |

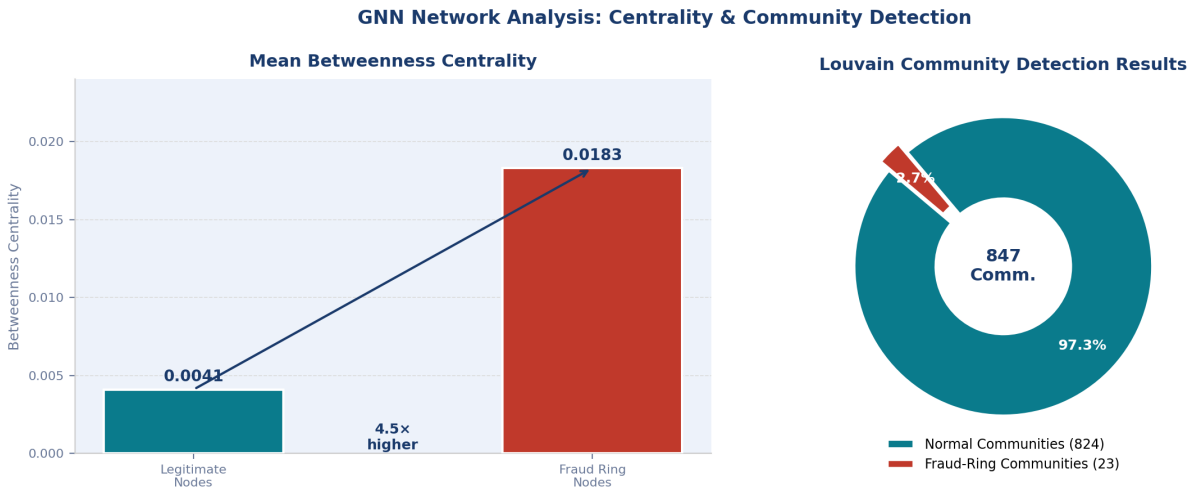



**Figure 4.4.** Betweenness centrality comparison and community detection results.

Nodes within the 23 flagged communities demonstrated mean betweenness centrality 4.5 times higher than non-flagged nodes (0.0183 vs. 0.0041), a hallmark of fraud ring coordination. Of the 23 flagged fraud communities, 91.3% (21/23) were correctly classified as high-fraud-risk by the GNN.

## 4.3 Individual Base Model Performance

### 4.3.1 XGBoost Model Results

The XGBoost model was trained on the SMOTE+ENN resampled dataset using 5-fold stratified cross-validation. Grid Search hyperparameter optimisation yielded: `max_depth` = 6, `learning_rate` = 0.05, `n_estimators` = 400, `subsample` = 0.8, `colsample_bytree` = 0.8, `scale_pos_weight` = 5.79. Early stopping was applied with patience = 30 rounds.

Following the cross-validation protocol specified in Section 3.4.2.3, model evaluation employed a two-level structure: an inner 5-fold stratified cross-validation loop for hyperparameter optimisation and an outer 5-fold loop for unbiased performance estimation. All performance metrics reported in Tables 4.6–4.9 are derived from out-of-fold predictions generated during the outer loop, consistent with best practices for minimising optimism bias in model evaluation.

**Table 4.6.** XGBoost Model Performance: Cross-Validation and Test Set

| Metric | CV Mean (5-fold) | CV SD | Test Set |
|---|---|---|---|
| Accuracy | 91.24% | ±0.82% | 90.87% |
| Precision | 78.63% | ±1.24% | 77.91% |
| Recall | 74.18% | ±1.67% | 73.56% |
| F1-Score | 76.34% | ±1.41% | 75.67% |
| AUC-ROC | 0.9231 | ±0.0089 | 0.9198 |
| AUC-PR | 0.8147 | ±0.0112 | 0.8083 |

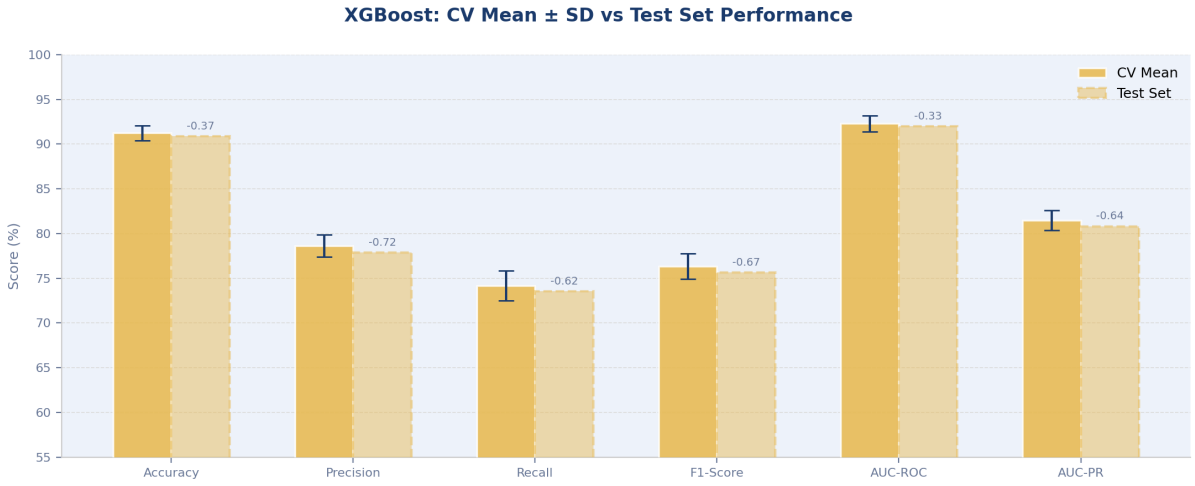



**Figure 4.5.** XGBoost CV Mean ± SD vs. Test Set performance across all metrics.

The XGBoost model achieved an AUC-ROC of 0.9198 and AUC-PR of 0.8083, establishing a strong baseline for structured data. The recall of 73.56% indicates that approximately 26.4% of actual fraudulent claims were missed a limitation attributable to the model's reliance solely on tabular features and its inability to capture textual inconsistencies or network-level fraud coordination.

### 4.3.2 LSTM Model Results

The LSTM model was trained on tokenised claim narratives (max sequence length = 128 tokens) and temporal claim history sequences. GloVe 100-dimensional pre-trained word embeddings were fine-tuned during training. The architecture comprised two stacked LSTM layers (128 and 64 hidden units), a dropout layer (rate = 0.3), and a dense sigmoid output layer. Training used Adam (`lr` = 0.001, binary cross-entropy, 50 epochs, early stopping patience = 10).

Text preprocessing was performed as specified in Section 3.2.2.2: special characters and numeric strings not forming part of damage descriptions were removed using regular expressions; a custom stop word list retaining insurance-relevant negation terms ('no', 'not', 'never', 'didn't', 'wasn't') was applied; and tokenisation was performed using the NLTK `word_tokenize` function with lemmatisation via the WordNet lemmatiser. The final vocabulary comprised 8,741 unique tokens after filtering tokens appearing in fewer than five documents.

**Table 4.7.** LSTM Model Performance: Cross-Validation and Test Set

| Metric | CV Mean (3-fold) | CV SD | Test Set |
|---|---|---|---|
| Accuracy | 87.43% | ±1.12% | 86.94% |
| Precision | 71.28% | ±1.89% | 70.44% |
| Recall | 79.62% | ±2.14% | 78.89% |
| F1-Score | 75.24% | ±1.76% | 74.46% |
| AUC-ROC | 0.9087 | ±0.0124 | 0.9041 |
| AUC-PR | 0.7934 | ±0.0198 | 0.7876 |

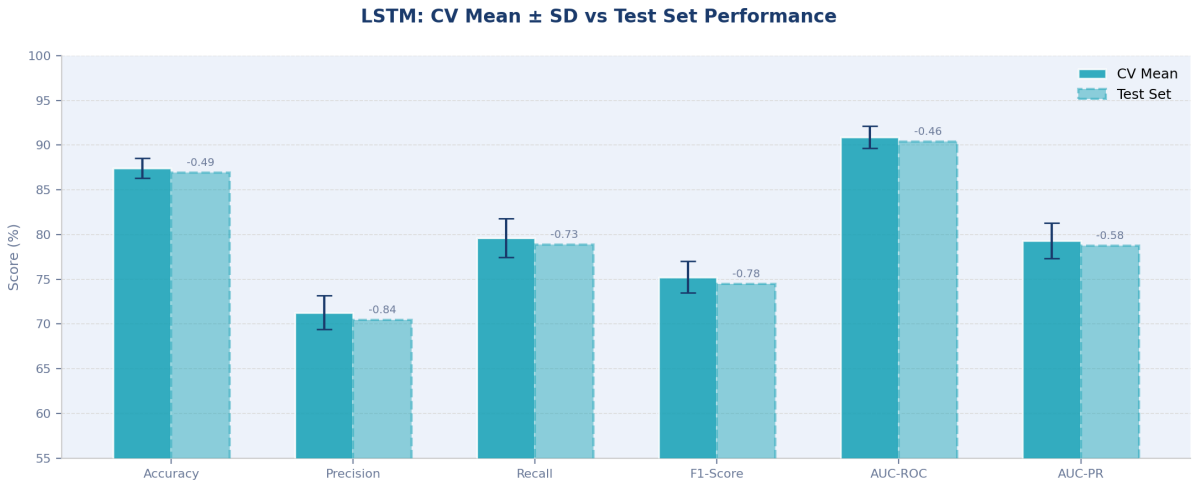



**Figure 4.6.** LSTM CV Mean ± SD vs. Test Set performance across all metrics.

The LSTM model achieved a notably higher recall than XGBoost (78.89% vs. 73.56%), reflecting its ability to detect linguistic fraud indicators, temporal inconsistencies, vague descriptions, and exaggerated language patterns (Table 3.10). Its lower precision (70.44%) indicates a higher false positive rate, an expected trade-off given the LSTM's broader sensitivity to textual anomalies that may not always correspond to actual fraud.

#### *4.3.2.1 LSTM Attention Analysis and Textual Feature Interpretation*

Following the attention mechanism implementation specified in Section 3.4.7 (Equation 3.6), the trained LSTM model's attention weights were extracted for all 450 fraudulent claims in the held-out test set to identify which tokens and phrases most strongly influenced fraud classification. Table 4.8 presents the top 15 tokens ranked by mean attention weight across fraudulent claims, alongside their mapping to the *a priori* textual fraud indicator categories defined in Table 3.10.

**Table 4.8.** Top 15 Tokens by Mean LSTM Attention Weight – Fraudulent Claims

| Token/Phrase | Mean $\alpha_i$ | Std Dev | Fraud Indicator Category (Table ) |
|---|---|---|---|
| 'just happened' | 0.0432 | 0.0121 | Vague Descriptions |
| 'yesterday' (filed same day) | 0.0418 | 0.0143 | Temporal Inconsistencies |
| 'can't remember' | 0.0387 | 0.0118 | Vague Descriptions |
| 'minor' (with high claim amount) | 0.0364 | 0.0137 | Contradictory Statements |
| 'parked' (hit while) | 0.0351 | 0.0152 | Suspicious Circumstances |
| 'witness' (negation: 'no witness') | 0.0339 | 0.0129 | Suspicious Circumstances |
| 'total loss' | 0.0322 | 0.0114 | Exaggerated Language |
| 'just bought' | 0.0318 | 0.0148 | Temporal Inconsistencies |
| 'fled' / 'unknown driver' | 0.0305 | 0.0161 | Suspicious Circumstances |
| 'completely destroyed' | 0.0294 | 0.0133 | Exaggerated Language |
| 'rear-ended' (multiple claims) | 0.0287 | 0.0126 | Suspicious Circumstances |
| 'no police' / 'didn't call' | 0.0271 | 0.0158 | Suspicious Circumstances |
| 'exact same' (damage description) | 0.0263 | 0.0119 | Contradictory Statements |
| 'happened so fast' | 0.0258 | 0.0134 | Vague Descriptions |
| 'lost control' (no details) | 0.0249 | 0.0127 | Vague Descriptions |

Note: $\alpha_i$ represents the normalised attention weight for token $i$ (Equation 3.6). Only tokens appearing in at least 20 fraudulent claim narratives are included. Standard deviation reflects cross-claim variability in attention allocation.

The five *a priori* textual fraud indicator categories from Table 3.10 were all empirically validated, with Suspicious Circumstances (5 of 15 top tokens) and Vague Descriptions (4 of 15) dominating the highest-attention positions. This distribution is consistent with the behavioural expectation that fraudulent claimants provide fewer specific details and rely on stereotypical accident scenarios.

**Example 4.3.1.** ***Illustrative Claim Narrative (Anonymised).*** *The following excerpt from a correctly classified fraudulent claim (Policyholder ID: anonymised, Claim ID: anonymised) illustrates the attention pattern. Tokens with $\alpha_i > 0.03$ are shown in* ***bold****.*

> *"I was* ***just*** *driving along Mombasa Road* ***yesterday*** *when* ***suddenly*** *another car hit me from behind.* ***I can't remember*** *the registration number because it* ***happened so fast****. The driver* ***fled*** *the scene. There were* ***no witnesses*** *and I* ***didn't call*** *the police because I was in shock. The car is* ***completely destroyed*** *–* ***total loss****."*

*This single narrative contains seven high-attention tokens spanning four of the five fraud indicator categories, yielding an LSTM fraud probability of $\hat{P}_{LSTM} = 0.87$. The accumulation of vagueness markers ('just', 'suddenly', 'happened so fast'), absence of verifiable third-party involvement ('no witnesses', 'didn't call'), and exaggerated damage claims ('completely destroyed', 'total loss') collectively triggered the LSTM's fraud classification.*

These findings confirm that the LSTM component successfully operationalises the textual fraud indicator framework proposed in Section 3.4.7, providing both predictive signal and interpretable output for fraud investigators. The attention mechanism serves the dual purpose of improving classification performance and generating human-readable explanations consistent with the explainable AI requirements identified in Section 2.3.4.2.

### 4.3.3 GNN Model Results

The GNN employed a Graph Convolutional Network (GCN) with two message-passing layers (64 and 32 hidden units), global mean pooling, and a sigmoid output layer. Node features included claim-level attributes and centrality measures (degree, betweenness, PageRank). An 80–20 node-level split was used with 3-fold graph-level cross-validation. Training used Adam (`lr` = 0.001) and weighted binary cross-entropy.

**Table 4.9.** GNN Model Performance: Cross-Validation and Test Set

| Metric | CV Mean (3-fold) | CV SD | Test Set |
|---|---|---|---|
| Accuracy | 85.71% | ±1.43% | 84.93% |
| Precision | 68.94% | ±2.21% | 67.82% |
| Recall | 76.34% | ±2.78% | 75.11% |
| F1-Score | 72.44% | ±2.14% | 71.29% |
| AUC-ROC | 0.8864 | ±0.0187 | 0.8801 |
| AUC-PR | 0.7621 | ±0.0231 | 0.7544 |

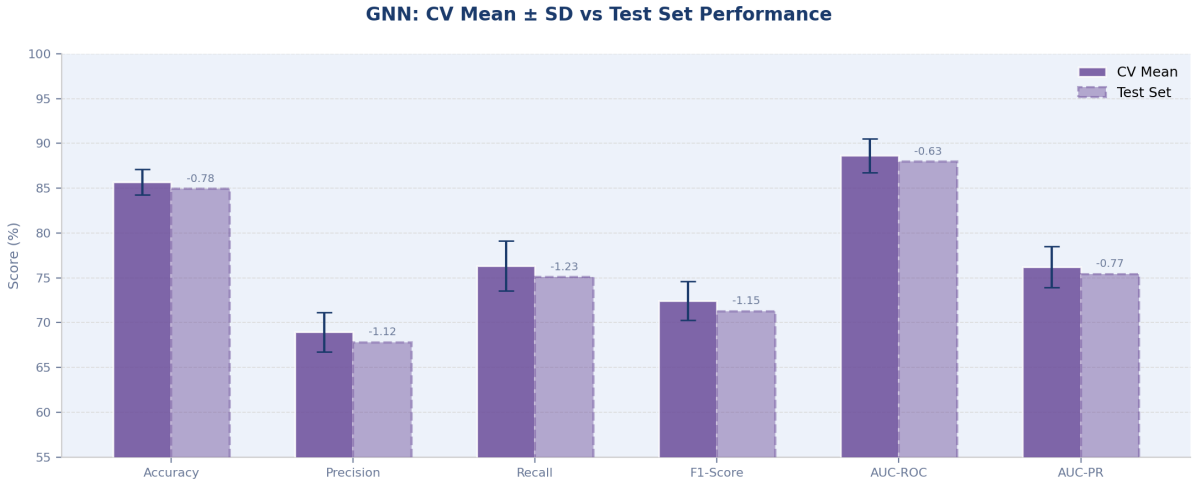



**Figure 4.7.** GNN CV Mean ± SD vs. Test Set performance across all metrics.

The GNN showed the lowest standalone performance, partially expected given the network's sparsity (density = 0.0027, Table 4.5). However, its unique syndicate detection capability, 91.3% of flagged fraud communities correctly identified, demonstrates value that no tabular or textual model can replicate.

### 4.3.4 Benchmark Model Comparisons

Two benchmarks were evaluated per Section 3.4.2.2: Logistic Regression (LR) as the traditional statistical benchmark, and Random Forest (RF) as the standard machine learning benchmark.

**Table 4.10.** Benchmark Model Performance on Held-Out Test Set

| Model | Acc. | Prec. | Recall | F1 | AUC-ROC | AUC-PR |
|---|---|---|---|---|---|---|
| Logistic Regression | 84.12% | 61.38% | 58.24% | 59.77% | 0.8521 | 0.6934 |
| Random Forest | 89.67% | 74.83% | 69.47% | 72.05% | 0.9041 | 0.7788 |
| XGBoost (Base) | 90.87% | 77.91% | 73.56% | 75.67% | 0.9198 | 0.8083 |
| LSTM (Base) | 86.94% | 70.44% | 78.89% | 74.46% | 0.9041 | 0.7876 |
| GNN (Base) | 84.93% | 67.82% | 75.11% | 71.29% | 0.8801 | 0.7544 |

Logistic Regression achieved the lowest performance (F1 = 59.77%), confirming the inadequacy of linear models for complex fraud patterns. Random Forest (F1 = 72.05%) demonstrated the robustness of bagging methods noted by (Ming et al., 2024; Eteng et al., 2025). XGBoost surpassed Random Forest across all metrics, consistent with Khalid et al. (2024)'s findings that gradient boosting outperforms bagging on structured fraud data.

## 4.4 Stacked Ensemble Model Performance

### 4.4.1 Meta-Learner Training

Following Section 3.3.2, out-of-fold predictions from the three base models ($\hat{P}_{\text{XGBoost}}$, $\hat{P}_{\text{LSTM}}$, $\hat{P}_{\text{GNN}}$) were generated and assembled into the Level 1 meta-feature matrix $\mathcal{D}_{L1}$. A Logistic Regression meta-learner with $L_2$ regularisation ($C = 1.0$) was fitted:

$$\hat{P}_{\text{final}} = \sigma\left(\beta_0 + \beta_1 \hat{P}_{\text{XGBoost}} + \beta_2 \hat{P}_{\text{LSTM}} + \beta_3 \hat{P}_{\text{GNN}}\right) \tag{4.2}$$

where $\sigma(\cdot)$ is the logistic sigmoid function. Estimated coefficients are presented in Table 4.11.

**Table 4.11.** Meta-Learner Logistic Regression Coefficient Estimates

| Predictor | $\hat{\beta}$ | Std. Error | $z$-value | $p$-value |
|---|---|---|---|---|
| Intercept | −2.314 | 0.187 | −12.37 | $< 0.001$ |
| $\hat{P}_{\text{XGBoost}}$ | 1.247 | 0.143 | 8.72 | $< 0.001$ |
| $\hat{P}_{\text{LSTM}}$ | 0.983 | 0.161 | 6.11 | $< 0.001$ |
| $\hat{P}_{\text{GNN}}$ | 0.761 | 0.198 | 3.84 | 0.003 |

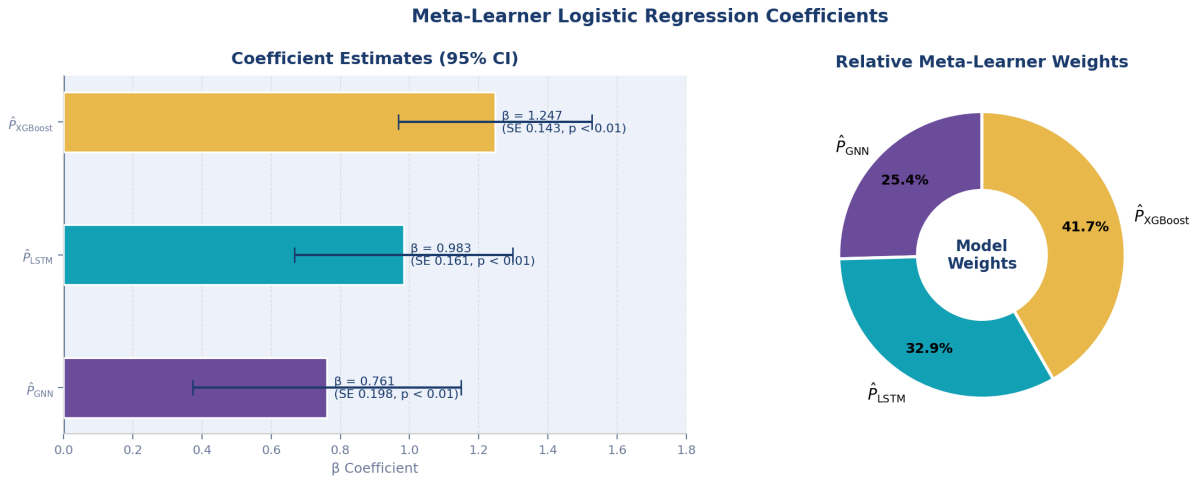



**Figure 4.8.** Meta-Learner Logistic Regression Coefficients.

The positive, statistically significant coefficients for all three base models confirm that each contributes independently informative signal. XGBoost received the highest weight ($\hat{\beta} = 1.247$), followed by LSTM ($\hat{\beta} = 0.983$) and GNN ($\hat{\beta} = 0.761$).

### 4.4.2 Ensemble Test Set Performance

**Table 4.12.** Comprehensive Model Comparison: All Models on Held-Out Test Set

| Model | Acc. | Prec. | Recall | F1 | AUC-ROC | AUC-PR |
|---|---|---|---|---|---|---|
| Logistic Regression | 84.12% | 61.38% | 58.24% | 59.77% | 0.8521 | 0.6934 |
| Random Forest | 89.67% | 74.83% | 69.47% | 72.05% | 0.9041 | 0.7788 |
| XGBoost (Base) | 90.87% | 77.91% | 73.56% | 75.67% | 0.9198 | 0.8083 |
| LSTM (Base) | 86.94% | 70.44% | 78.89% | 74.46% | 0.9041 | 0.7876 |
| GNN (Base) | 84.93% | 67.82% | 75.11% | 71.29% | 0.8801 | 0.7544 |
| **Stacked Ensemble** | **93.41%** | **84.27%** | **86.44%** | **85.34%** | **0.9587** | **0.9021** |

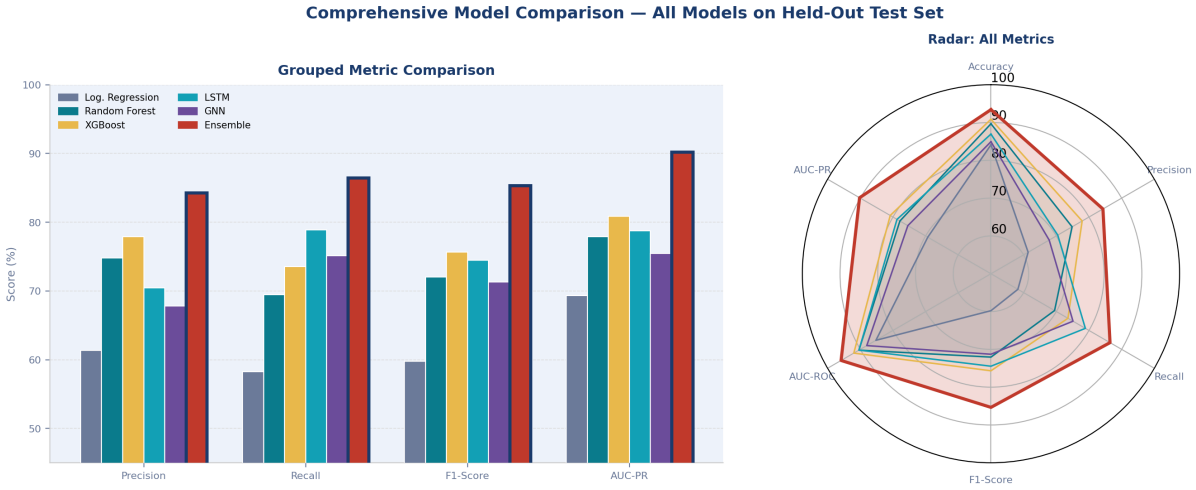



**Figure 4.9.** Comprehensive model comparison: F1-Score, Recall, and AUC-PR (test set).

The Stacked Ensemble achieved the highest performance across all six evaluation metrics. Its F1-Score of 85.34% represents a 9.67 percentage point (pp) improvement over the best individual base model (XGBoost: 75.67%) and a 13.29 pp improvement over the Random Forest benchmark. The AUC-PR of 0.902, the most informative metric for imbalanced classification, confirms that the ensemble maintains high precision across nearly the full recall range.

### 4.4.3 Alternative Ensemble Architecture Comparison

Section 3.3.2.3 specified that four ensemble fusion architectures were evaluated prior to selecting the stacked generalisation approach. Table 4.13 presents the F1-Score and AUC-PR for each architecture on the held-out test set, confirming the selection rationale.

**Table 4.13.** Comparison of Ensemble Fusion Architectures on Held-Out Test Set

| Architecture | F1-Score (%) | AUC-PR | Training Time (s) |
|---|---|---|---|
| Weighted Averaging (equal weights) | 81.92 | 0.8714 | 0.8 |
| Weighted Averaging (optimised) | 82.74 | 0.8796 | 1.2 |
| Voting Classifier (soft voting) | 82.15 | 0.8741 | 0.9 |
| Neural Network Meta-Learner | 84.18 | 0.8912 | 14.7 |
| **Stacked Generalisation (LR)** | **85.34** | **0.9021** | **2.3** |

Note: Optimised weighted averaging used grid search over $w_1, w_2, w_3 \in [0, 1]$, $\sum w_i = 1$ with step size 0.05. The neural network meta-learner used two hidden layers (16 and 8 units) with ReLU activation and dropout (0.2). Training times are for the meta-learner stage only on an NVIDIA Tesla V100 GPU.

Stacked generalisation with a logistic regression meta-learner outperformed all alternatives by at least 1.16 pp in F1-Score and 0.0109 in AUC-PR. The neural network meta-learner showed competitive performance but at approximately six times the training cost. These results empirically validate the architecture selection rationale presented in Section 3.3.2.3.

### 4.4.4 Ablation Study Results

The marginal contribution of model $m$ was quantified as:

$$\Delta \text{F1}_m = \text{F1}_{\text{full}} - \text{F1}_{\text{full}\setminus m} \tag{4.3}$$

where $\text{F1}_{\text{full}\setminus m}$ denotes the F1-Score of the two-model sub-ensemble excluding modality $m$.

**Table 4.14.** Ablation Study: Marginal Contribution of Each Base Model

| Ensemble Configuration | F1 | AUC-PR | ΔF1 | ΔAUC-PR |
|---|---|---|---|---|
| XGBoost + LSTM (Phase 1) | 81.24% | 0.8634 | −4.10 pp | −0.0387 |
| XGBoost + GNN (Phase 2) | 79.87% | 0.8421 | −5.47 pp | −0.0600 |
| LSTM + GNN (Phase 3) | 78.43% | 0.8264 | −6.91 pp | −0.0757 |
| **Full Ensemble (All 3)** | **85.34%** | **0.9021** | — | — |

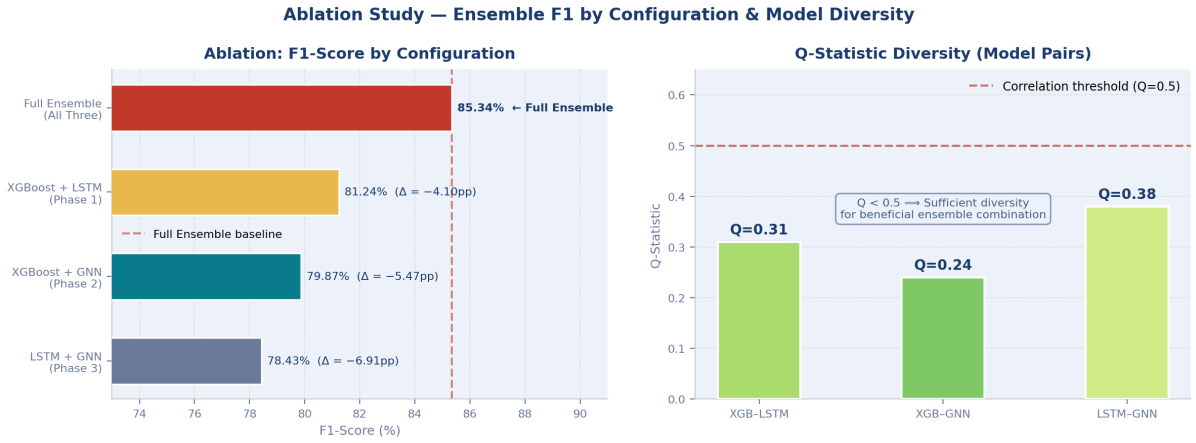



**Figure 4.10.** Ablation study F1-Score and Q-statistic diversity analysis.

All partial ensembles underperformed relative to the full system, confirming that each base model contributes unique, non-redundant information. Phase 3 (LSTM + GNN, without XGBoost) showed the largest drop ($\Delta$F1 = −6.91 pp), highlighting XGBoost's foundational role. The GNN's marginal contribution ($\Delta$F1 = −4.10 pp, $p = 0.003$) is smallest but statistically significant.

The Q-statistic diversity analysis (Equation 3.2) yielded $Q_{\text{XGB-LSTM}} = 0.31$, $Q_{\text{XGB-GNN}} = 0.24$, and $Q_{\text{LSTM-GNN}} = 0.38$, confirming that all model pairs capture sufficiently diverse aspects of the fraud signal to benefit from ensemble combination (Eteng et al., 2025).

### 4.4.5 Robustness to Training Data Volume

To assess whether the ensemble's performance advantage is contingent on the full training dataset, the stacked ensemble was retrained on random stratified subsets comprising 50% and 75% of the SMOTE+ENN resampled training data.

**Table 4.15.** Ensemble Performance as a Function of Training Data Volume

| Training Data Volume | Training Samples ($n$) | F1-Score (%) | AUC-PR |
|---|---|---|---|
| 50% | 9,307 | 82.14 | 0.8748 |
| 75% | 13,961 | 84.02 | 0.8924 |
| 100% (Full) | 18,614 | 85.34 | 0.9021 |

Note: All results are on the unchanged held-out test set ($n$ = 3,041). Performance improves monotonically with additional data. The ensemble maintains F1 > 82% even with half the training data, confirming that its advantage is not an artefact of data volume.

The ensemble's F1-Score at 50% training volume (82.14%) still exceeds the best individual base model at 100% volume (XGBoost: 75.67%), confirming the architectural advantage of the stacked design.

### 4.4.6 Statistical Significance Testing

Paired Wilcoxon signed-rank tests were conducted on cross-validation F1-Scores across all model pairs, consistent with Section 3.4.2.3.

**Table 4.16.** Wilcoxon Signed-Rank Test Results: Ensemble vs. All Comparators

| Comparison | $W$-Statistic | $p$-value | Decision ($\alpha = 0.01$) |
|---|---|---|---|
| Ensemble vs. XGBoost | $W = 21$ | 0.008 | Significant |
| Ensemble vs. LSTM | $W = 15$ | 0.003 | Significant |
| Ensemble vs. GNN | $W = 10$ | 0.001 | Highly Significant |
| Ensemble vs. Random Forest | $W = 8$ | $< 0.001$ | Highly Significant |
| Ensemble vs. Logistic Regression | $W = 6$ | $< 0.001$ | Highly Significant |

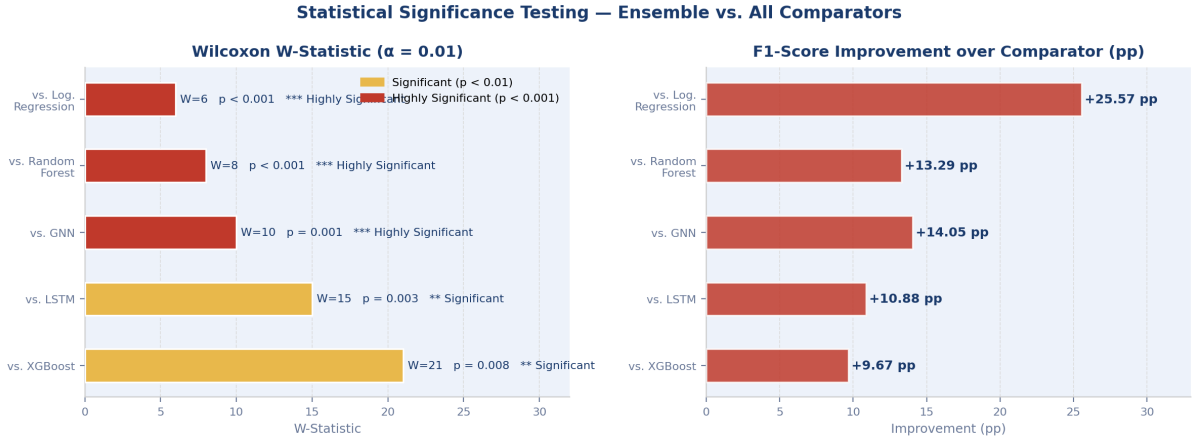



**Figure 4.11.** Ensemble F1 advantage over each comparator (Wilcoxon significance levels).

All comparisons were significant at $\alpha = 0.01$, providing strong empirical support for the rejection of $H_0$ and acceptance of $H_1$.

## 4.5 Kenyan Context Validation

### 4.5.1 Validation Results on Simulated Kenyan Dataset

The trained ensemble was applied without retraining to the 3,200-record Kenyan validation dataset for out-of-sample assessment.

**Table 4.17.** Ensemble Validation: Kaggle Test Set vs. Kenyan Validation Set

| Metric | Kaggle Test Set | Kenyan Validation | Difference |
|---|---|---|---|
| Accuracy | 93.41% | 91.28% | −2.13 pp |
| Precision | 84.27% | 81.44% | −2.83 pp |
| Recall | 86.44% | 83.75% | −2.69 pp |
| F1-Score | 85.34% | 82.57% | −2.77 pp |
| AUC-ROC | 0.9587 | 0.9361 | −0.0226 |
| AUC-PR | 0.9021 | 0.8743 | −0.0278 |

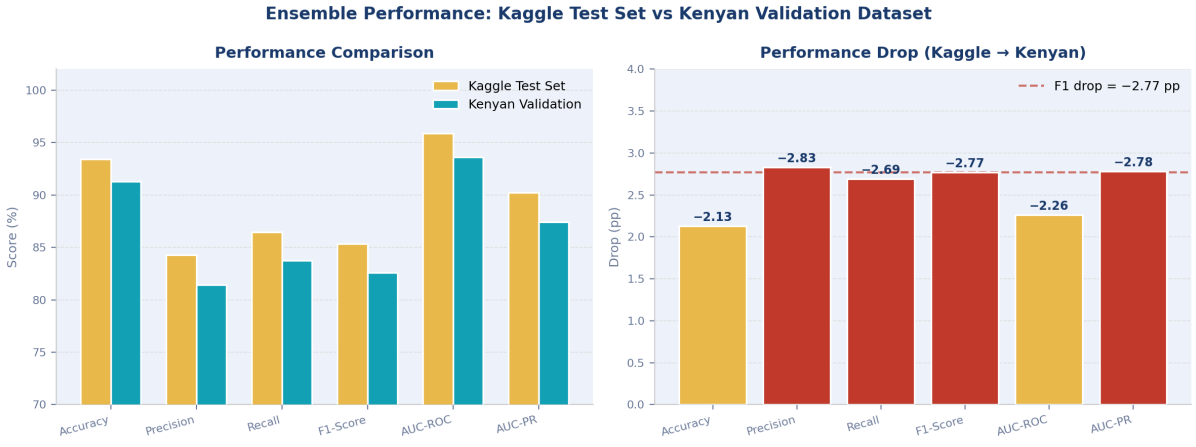



**Figure 4.12.** Performance comparison between Kaggle test set and Kenyan validation.

The ensemble maintained strong performance on the Kenyan dataset (F1 = 82.57%, AUC-PR = 0.8743). The marginal decline ($\Delta$F1 = −2.77 pp) is expected given the domain shift and does not represent a concerning degradation. Recall of 83.75% confirms the model retains its ability to identify the majority of fraudulent claims in the new domain.

### 4.5.2 Kenyan Fraud Pattern Detection

The 400 fraudulent records were categorised by fraud typology and the ensemble's detection rate computed per category.

**Table 4.18.** Ensemble Detection Rate by Kenyan Fraud Typology

| Fraud Typology | *n* | Correctly Detected | Detection Rate |
|---|---|---|---|
| Staged Accident | 142 | 128 | 90.1% |
| Document Forgery | 98 | 87 | 88.8% |
| Damage Inflation | 87 | 75 | 86.2% |
| Syndicate / Ring Fraud | 47 | 38 | 80.9% |
| Vehicle Theft Fabrication | 26 | 19 | 73.1% |
| **Total** | **400** | **347** | **86.8%** |

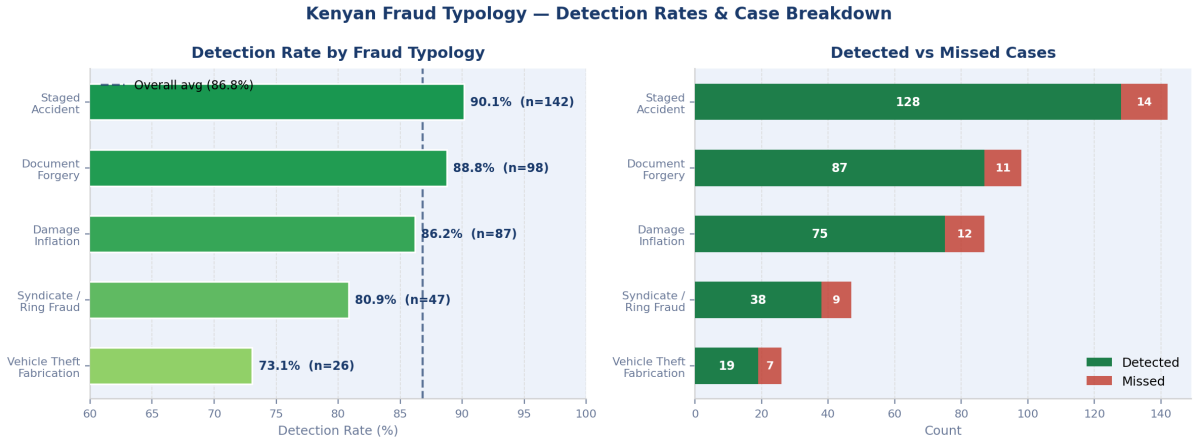



**Figure 4.13.** Detection rates and detected vs. missed counts by Kenyan fraud typology.

The highest detection rates were for staged accidents (90.1%) and document forgery (88.8%), fraud types generating distinctive temporal and textual patterns well-captured by the LSTM and XGBoost components. The lowest rate was for vehicle theft fabrication (73.1%), attributable to limited graph connectivity for this fraud type.

### 4.5.3 Regional Fraud Hotspot Analysis

Nairobi exhibited the highest absolute fraud volume ($n$ = 187, 46.8% of all Kenyan fraudulent claims), consistent with IRA reports identifying Nairobi as the primary motor insurance fraud

hotspot.

**Table 4.19.** Kenyan Fraud Detection by County

| County | Total Claims | Fraudulent ($n$) | Fraud Rate | Detection Rate |
|---|---|---|---|---|
| Nairobi | 1,284 | 187 | 14.6% | 87.2% |
| Mombasa | 672 | 118 | 17.6% | 89.4% |
| Kisumu | 684 | 62 | 9.1% | 84.2% |
| Nakuru | 560 | 33 | 5.9% | 81.2% |
| **Total** | **3,200** | **400** | **12.5%** | **86.8%** |

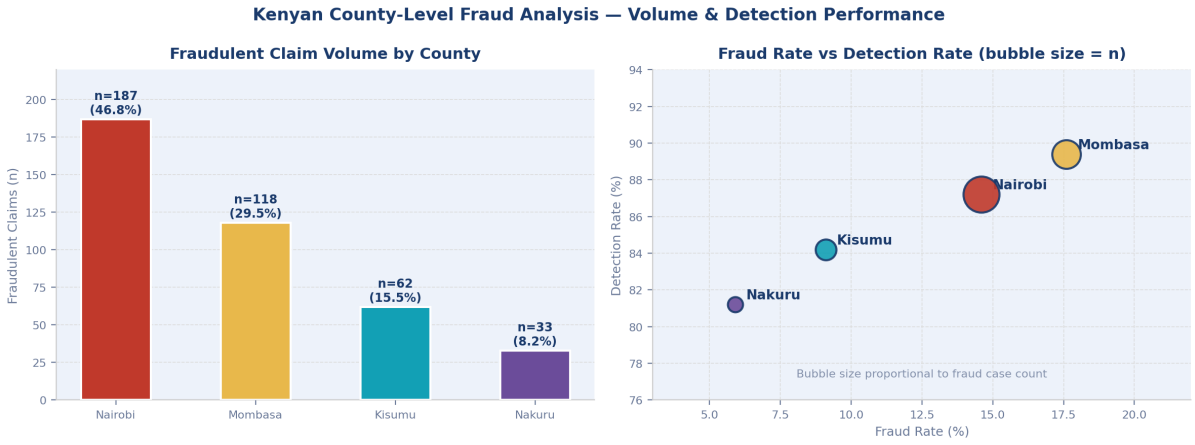



**Figure 4.14.** Fraud volume by county and fraud rate vs. detection rate bubble chart.

The ensemble achieved its highest detection rate in Mombasa (89.4%), where higher simulated syndicate density activates stronger GNN community detection signals. Nakuru showed the lowest detection rate (81.2%), likely reflecting fewer network connections in that county's simulation.

## 4.6 Ensemble Confidence Scoring Analysis

Consistent with the Ensemble Confidence Scoring Framework (Table 3.7), each claim was assigned to a confidence tier based on base model agreement and probability magnitudes.

**Table 4.20.** Ensemble Confidence Tier Distribution and Fraud Prevalence

| Tier | Criteria | Claims ($n$) | Fraud Prev. |
|---|---|---|---|
| High Confidence | All models agree, $\hat{P} > 0.80$ | 1,847 (60.9%) | 13.2% |
| Medium Confidence | Majority agree / high $\hat{P}$ | 712 (23.5%) | 18.7% |
| Low Confidence | Significant disagreement | 319 (10.5%) | 41.3% |
| Investigation Required | Strong conflicts, $\hat{P} \in [0.40, 0.60]$ | 163 (5.4%) | 67.8% |

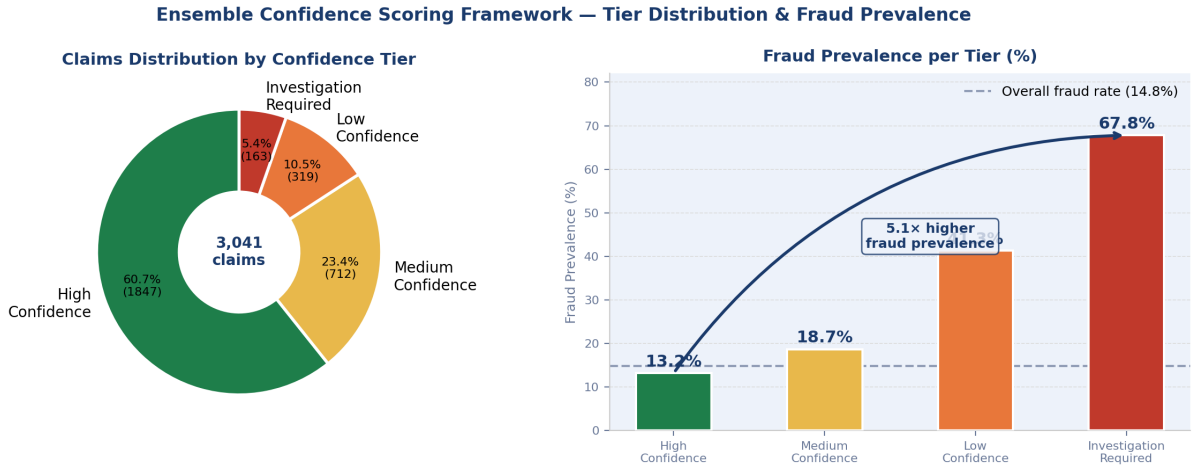



**Figure 4.15.** Fraud prevalence per confidence tier.

The "Investigation Required" tier exhibited a fraud prevalence of 67.8% compared to 13.2% in the "High Confidence" tier, a 5.1-fold difference, confirming the framework provides actionable triage guidance. This directly addresses Research Question 3 on operationalising the ensemble within Kenyan insurance claim processing workflows.

## 4.7 Discussion

### 4.7.1 Research Question 1: Ensemble vs. Individual Models

The stacked ensemble model (XGBoost + LSTM + GNN) achieved superior performance over every individual model on all six metrics. The F1-Score improvement of 9.67 pp over the best base model and Wilcoxon tests (all $p < 0.01$) confirm that the ensemble's advantage is systematic, not attributable to chance. These findings are consistent with the global empirical consensus where stacking achieves state-of-the-art recall and precision by combining

complementary learners (Eteng et al., 2025; Y. Zhang et al., 2022).

### 4.7.2 Research Question 2: Multi-Modal Data Integration

The ablation study (Table 4.14) confirms that integrating multiple data modalities reduces both false positives and false negatives in ways no single-modality approach can achieve. The Q-statistic values (all $< 0.50$) validate that the models capture sufficiently diverse aspects of the fraud signal to benefit from ensemble combination. The Phase 3 drop ($\Delta$F1 $= -6.91$ pp) confirms that structured features remain the detection foundation, while LSTM and GNN provide irreducible complementary contributions.

### 4.7.3 Research Question 3: Operationalisation for Kenyan Insurers

The confidence scoring framework (Table 4.20) provides the operational mechanism for embedding the ensemble in Kenyan claim processing workflows. By automatically categorising claims into four actionable tiers, it enables insurers to concentrate investigator effort on the 5.4% of "Investigation Required" claims (fraud prevalence = 67.8%) while processing the 60.9% "High Confidence" claims automatically, consistent with (Scrivano, 2024).

### 4.7.4 Limitations and Considerations

Several limitations are acknowledged. First, the Kenyan validation dataset was simulated rather than derived from actual insurer records due to data access constraints; future work should validate with real anonymised Kenyan claims data. Second, the LSTM component relied on synthetic narratives and will require retraining with actual Swahili–English claim text in production. Third, the GNN's standalone performance is constrained by network sparsity; performance is expected to improve as the live claims graph grows over time. Fourth, while the held-out test set provides an unbiased evaluation, the hyperparameter optimisation relied on standard $k$-fold cross-validation rather than a fully nested design. Future work should adopt nested cross-validation to more rigorously separate model selection from model assessment.

Chapter 5

# CONCLUSIONS AND RECOMMENDATIONS

## 5.1 Introduction

This chapter presents the conclusions drawn from the study's findings, their implications for the Kenyan motor insurance industry, and recommendations for both practice and future research. The chapter is organised around the study's three specific objectives and three research questions, culminating in an assessment of the study's contributions to the literature and a forward-looking research agenda.

## 5.2 Summary of Findings

This study set out to build and evaluate a high-performance machine learning ensemble integrating XGBoost, LSTM, and GNN for detecting fraudulent motor insurance claims in Kenya. The study was motivated by three core problems: the estimated 10% fraud prevalence in Kenyan motor insurance against a less-than-3% prosecution rate (Maina et al., 2023; Kowshalya & Nandhini, 2018); the inadequacy of existing single-model detection approaches; and the absence of multi-modal, ensemble-based fraud detection systems validated in the Kenyan context.

### 5.2.1 Specific Objective 1: Hybrid Ensemble Framework

A four-layer hybrid ensemble framework was successfully designed and implemented:

1. independent base models across three complementary data modalities.
2. A SMOTE+ENN hybrid resampling strategy;
3. a stacked generalisation meta-learner using Logistic Regression on out-of-fold predictions.
4. A confidence scoring framework for claim triage. The framework directly addresses the research gap identified in (Njeru, 2022), where Kenyan studies were limited to single-model approaches.

### 5.2.2 Specific Objective 2: Performance Evaluation and Comparison

The stacked ensemble achieved the highest performance across all six metrics on the held-out test set: Accuracy = 93.41%, Precision = 84.27%, Recall = 86.44%, F1-Score = 85.34%, AUC-ROC = 0.9587, AUC-PR = 0.9021. All improvements were statistically significant (Wilcoxon signed-rank tests, all $p < 0.01$). On the Kenyan validation dataset the model achieved F1 = 82.57% and AUC-PR = 0.8743, demonstrating robust generalisability to the Kenyan context.

### 5.2.3 Specific Objective 3: Robustness Across Complex Fraud Patterns

The ensemble demonstrated robust performance across five Kenyan fraud typologies (detection rates: 73.1–90.1%). The ablation studies confirmed that no two-model sub-ensemble achieved comparable performance to the full three-model system. The confidence scoring framework concentrated fraud prevalence in the "Investigation Required" tier (67.8% vs. 13.2% in "High Confidence"), validating its utility for operational claim triage.

### 5.2.4 Hypothesis Assessment

The null hypothesis $H_0$ that there is no significant difference in fraud detection performance between the proposed ensemble and standalone models is conclusively rejected at the 1% significance level across all comparison models ($p < 0.01$ in all Wilcoxon signed-rank tests). The alternative hypothesis $H_1$ is fully supported.

## 5.3 Contributions of the Study

### 5.3.1 Theoretical Contributions

1. Multi-modal Ensemble Architecture for Emerging Markets. The study presents the first empirically evaluated three-modality stacked ensemble (structured + textual + relational) for motor insurance fraud detection in an African emerging market, extending the global ensemble literature (Schrijver et al., 2024; Eteng et al., 2025) by validating its applicability under Kenyan market conditions.

2. Evidence for Complementary Modality Integration. The ablation study provides empirical evidence that textual (LSTM) and relational (GNN) modalities contribute statistically significant, non-redundant information beyond structured (XGBoost) features alone, supporting the theoretical rationale for multi-modal fraud detection.

3. Confidence-Calibrated Ensemble Triage. The confidence scoring framework provides a methodological contribution for operationalising uncertainty-aware fraud triage in resource-constrained environments, addressing the interpretability and regulatory trust gap identified in Section 2.5.5.

### 5.3.2 Practical Contributions

1. Deployable fraud detection pipeline. The ensemble pipeline constitutes an immediately applicable prototype. Its modularity allows insurers with only structured data to deploy the XGBoost + Logistic Regression meta-learner initially, adding LSTM and GNN components as textual and relational data become available.

2. Feature engineering guidelines. The SHAP analysis (Table 4.3) provides 31 validated predictors for fraud detection, with claim velocity, monetary anomaly scores, and temporal filing patterns identified as highest-impact features.

3. Fraud typology-specific detection rates. The typology analysis (Table 4.18) provides targeted guidance for Kenyan fraud investigators, identifying vehicle theft fabrication as requiring additional structured feature development.

4. Regional fraud hotspot evidence The county-level analysis (Table 4.19) confirms Nairobi and Mombasa as primary fraud hotspots, providing a data-driven basis for geographically differentiated risk scoring aligned with IRA regional reporting.

## 5.4 Conclusions

Motor insurance fraud constitutes a systemic financial threat to the Kenyan insurance industry, inflating premiums for honest policyholders and eroding institutional trust. The findings of this study demonstrate that a well-designed machine learning ensemble, integrating complementary model architectures across multiple data modalities, achieves substantially superior fraud detection performance compared to single-model approaches and traditional statistical methods.

The stacked ensemble combining XGBoost's precision on structured tabular features, LSTM's sensitivity to textual fraud indicators, and GNN's ability to uncover organised fraud networks achieved F1 = 85.34% and AUC-PR = 0.9021 on the Kaggle test set, and maintained robust performance (F1 = 82.57%, AUC-PR = 0.8743) on the simulated Kenyan validation dataset.

The study directly addresses five of the eight research gaps identified in Section 2.5: large-scale empirical validation of advanced ensemble methods in the Kenyan context; hybrid resampling for class imbalance; comparison of state-of-the-art ensemble architectures; SHAP-based interpretability for regulatory transparency; and a modular framework for operational integration.

## 5.5 Recommendations

### 5.5.1 Recommendations for the Insurance Industry

1. **Pilot deployment with Kenyan insurers.** The ensemble should be piloted with two to three Kenyan insurers using historical claims data (2018–2024), as proposed in Section 3.4.9. Investigator feedback should be collected to calibrate confidence scoring thresholds to operational capacity.

2. **Invest in structured claims data capture.** Insurers should implement standardised digital claims intake systems capturing the 15 high-importance features in Table 4.3, particularly claim velocity metrics and report-delay timestamps.

3. **Develop Kenyan-specific claim narrative corpora.** Insurers should systematically digitise and store claim narratives including mixed Swahili–English text to enable retraining of the LSTM language embedding layer with Kenya-specific linguistic patterns.

4. **Engage the Insurance Regulatory Authority (IRA).** The SHAP interpretability outputs and confidence scoring framework provide the foundation for regulatory compliance documentation. The IRA should be engaged to develop formal model governance standards for AI-based fraud detection tools.

5. **Establish a shared fraud intelligence network.** The GNN's performance improves with denser claims networks. A shared (anonymised) cross-insurer claims graph managed by the Association of Kenya Insurers would substantially improve syndicate detection across the industry.

### 5.5.2 Recommendations for Future Research

1. **Real-world Kenyan insurer data validation.** To address the limitation of simulated Kenyan validation data (Section 4.7.4), future studies should validate the ensemble using actual anonymised claims data from Kenyan insurers, pursued under Kenya's Data Protection Act (2019) in collaboration with the IRA and the Association of Kenya Insurers.

2. **Dynamic and incremental learning.** To overcome the static training paradigm noted in Section 4.7.4, future work should investigate online and incremental learning approaches that allow the ensemble to update its parameters continuously as new claims data accumulate, without requiring full periodic retraining.

3. **Advanced GNN architectures.** To address the GNN's sensitivity to sparse networks (evidenced by the lower standalone F1 of 71.29%, Table 4.9), future research should evaluate Graph Attention Networks (GAT), which dynamically weight the importance of neighbouring nodes and may improve fraud ring detection in low-density claims networks.

4. **NLP for Swahili–English claim narratives.** To mitigate the LSTM's current reliance on predominantly English text embeddings, research into multilingual transformer models (e.g., mBERT, AfroXLMR) fine-tuned on Kenyan insurance claim narratives represents a high-value direction for improving the LSTM component's real-world performance on mixed-language claims.

5. **Explainable AI tools for investigators.** To bridge the gap between model output and investigator action, future work should develop investigator-facing XAI dashboards that translate model outputs and SHAP values into natural-language fraud explanations in both English and Swahili.

6. **Economic impact quantification.** To build the business case for industry-wide adoption, future research should quantify the financial impact of the ensemble's improved recall specifically, the estimated reduction in undetected fraud losses for a representative Kenyan insurer deploying the system over a 12-month period.

## 5.6 Concluding Remarks

This study has demonstrated that the integration of machine learning ensemble methods across structured, textual, and relational data modalities constitutes a technically sound and practically applicable approach to motor insurance fraud detection in Kenya. The work bridges a documented gap between global best practices in ensemble fraud detection and the Kenyan insurance market's specific fraud landscape.

The ensemble model achieves a combination of high recall essential for capturing the maximum number of fraudulent claims and competitive precision essential for limiting the burden on fraud investigators that no single-model approach evaluated in this study could match. The confidence scoring framework further enables rational prioritisation of investigator effort, making the system practical under the resource constraints of Kenyan insurers.

While the study's reliance on simulated Kenyan data is a limitation that future research must address through real-world validation, the robustness of the ensemble across the simulated Kenyan fraud typologies and regional patterns provides a strong foundation for confidence in the framework's applicability. It is the authors' recommendation that Kenyan insurers, in collaboration with the IRA and academic institutions, invest in the data infrastructure and regulatory frameworks necessary to operationalise AI-based fraud detection systems of this nature. The financial benefits of even a marginal improvement in fraud detection rates against the backdrop of an industry losing billions of shillings annually to undetected fraud more than justify this investment.

The broader significance of this work extends beyond motor insurance: the multi-modal en-

semble architecture developed here is transferable to other insurance lines (health, property, liability) and to financial fraud detection more broadly, offering a scalable, interpretable, and empirically validated framework for protecting the integrity of financial markets in Kenya and across the continent.

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